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Experimental investigation and thermodynamic assessment of the Al-Fe system

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ABSTRACT

The heat capacity of ε (Fe_5Al_8) was determined between 1395 and 1455 K by DSC measurements. The homogeneity ranges of the intermetallic phases ζ ($FeAl_2$), η (Fe_2Al_5) and θ (Fe_4Al_{13}) in the temperature range between 1073 and 1435 K were determined using WDS measurements on annealed and quenched samples in the composition range between 56.0 and 75.3 at.% Al. The solidus and liquidus temperatures of all samples were determined using DTA. A CALPHAD-type assessment of the Al-Fe system was done based on latest experimental results on heat capacity of the intermetallic phases.

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1. Introduction

The knowledge of phase relations in the Al-Fe system is important to understand the properties and microstructure of commercial Al-alloys, which contain also other components such as Si and Mg. Even small amounts of Fe may cause formation of intermetallic compounds substantially influencing mechanical properties of Al-alloys [1,2]. The Al-Fe system is therefore a key system in thermodynamic modelling to investigate the phase relations of such alloys.

A detailed review of experimental studies of the Al-Fe system can be found in Ref. [3] (see Fig. 1 for their assessed phase diagram) and several new experimental works in Refs. [4–13]. Thermodynamic parameters of the Al-Fe system were reported in Refs. [3,14–18].

The homogeneity ranges of the Al-rich intermetallic phases were studied in Refs. [6,7]. In both works, the shifting of the homogeneity ranges of ζ ($FeAl_2$), η (Fe_2Al_5) and θ (Fe_4Al_{13}) to more Fe-rich compositions with increasing temperature was reported. Differences can be seen in the behaviour of the θ-phase solubility range with temperature. According to [6], θ becomes more Fe-rich up to a temperature around 1273 K. Above this temperature, the solubility range narrows again and its Fe-content decreases. In comparison, the solubility range of θ is shifting to more Fe-rich compositions up to the peritectic melting point according to [7].

Heat capacities of the intermetallic phases ζ, η and θ were reported in Refs. [5,8,12,13]. The heat capacity of the high temperature ε phase (Fe_5Al_8) was not determined so far.

In the works presenting a CALPHAD assessment [3,14,15,17,18], the heat capacities were estimated using the additive Neumann-Kopp rule. Substantial drawback of the Neumann-Kopp approach in combination with SGTE unary database [19] is that effects of phase transformations occurring in elements will be present in the thermodynamic functions of an intermetallic as well [15]. For example, the change of c_p due to melting of aluminium will be observed in the c_p curve of aluminium-containing intermetallic compounds which melt at substantially higher temperature than pure Al.

The aim of the present work was to determine the heat capacity of ε experimentally. Similar to the work reported in Refs. [6,7], the solubility ranges of the Al-rich intermetallics and the phase transition temperatures of Al-rich samples were studied. Based on the experimental results of this study and the reported heat capacity of

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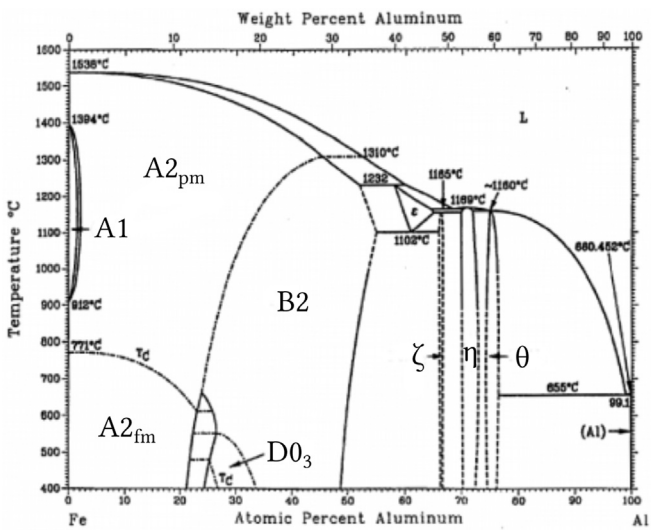

Fig. 1. The assessed Fe-Al phase diagram reported by Kattner and Burton (1993) [3].

the intermetallic phases ζ, η and θ [8,12], a new thermodynamic assessment of the Al-Fe system was done without using the Neumann-Kopp approach for modelling the intermetallic phases.

2. Methods

2.1. Sample preparation and characterisation

Fe-Al alloys with compositions between 56.0 and 75.3 at.% Al were prepared by arc melting (AM furnace, Edmund Bühler GmbH, Hechingen, Germany) using pure metals (Al 99.9999; Fe 99.995; Alfa Aesar, Karlsruhe, Germany). Sample compositions are given as nominal ones in this work. The as-cast samples were annealed under argon atmosphere at different temperatures between 1073 and 1426 K. For the lowest temperature, the samples were heat treated for at least 42 h. For the higher temperatures, the samples were heat treated between 7 h and 21 h, depending on composition and the expected phase assemblage. From evaluation of the corresponding microstructures, equilibrium was achieved in all samples. After annealing, the samples were quenched into water. The furnace was temperature calibrated to achieve the nominal temperature within ±5 K.

Transition temperatures of each sample were determined using DTA (Setaram SetSys Evolution 1750, Setaram, France). The DTA device was temperature calibrated by determining the melting points of Al, Ag, Cu and Au. In all experiments, the samples were heated in alumina crucibles under flowing argon atmosphere $\left(20\,\mathrm{ml}/_{\mathrm{min}}\right)$. The as-cast samples were equilibrated for at least 30 min below the solidus temperature before each measurement.

The transition temperature was measured by determining the onset point (OP) as the intersection of the extrapolated, unaffected baseline with the tangent of the peak of the measured DTA signal. Transition temperatures were determined for heating rates of 10, 5 and 2 $\mathrm{K}/_{\mathrm{min}}$. The heating rate dependency of the onset point temperatures were examined by a linear fit of these data. From that, the temperature of each transition was taken as the calculated onset point at a heating rate of 0 $\mathrm{K}/_{\mathrm{min}}$.

After quenching and DTA experiments, the samples were grinded and polished and were studied using SEM/EDS (Leo1530 Gemini, Carl Zeiss with EDS from Bruker AXS Microanalysis GmbH) in back-scattered electron contrast mode and SEM/EBSD (Nordlys II EBSD detector, Oxford Instruments).

2.2. EPMA/WDS

If possible, the chemical composition of phases of the quenched samples were measured by electron probe microanalysis with wavelength-dispersive X-ray spectroscopy (EPMA/WDS) using a JXA8900 RL (Jeol GmbH, Germany). The device was calibrated against standards of pure aluminium and iron. A minimum of 50 points for each phase was measured using line measurements with a stepsize of 0.5 μm. The obtained composition values were then size sorted and plotted as shown in Fig. 2 on the example of the two-phase sample ζ + η of the Al-Fe system (67.0 at.% Al) annealed at a temperature of 1073 K. If a distinct composition plateau was observed, the mean value and the corresponding standard deviation was calculated from an appropriate range of points for each of it. The determined standard deviation was in most cases below 0.2 at.%.

2.3. Heat capacity measurements

The heat capacity of the 60.9 at.% Al sample was measured in the temperature range between 1395 and 1455 K in a platinum crucible with an alumina inlay and a platinum lid under flowing argon atmosphere $\left(20\ \mathrm{ml}/_{\mathrm{min}}\right)$ using a NETZSCH Pegasus 404C DSC (NETZSCH-Gerätebau GmbH, Selb, Germany). The same setup was used for the previously reported c_p measurements in the Al-Fe system [8,12]. According to DTA (see section 3.1.1), the sample can be assumed to be single phase ε (Fe_5Al_8) in that temperature range. It is not possible to retain the ε phase in quenched samples as it will always transform eutectoidicaly to B2 + ζ on cooling.

The values of the heat capacity will be given in per mole of atoms in this work.

2.4. Thermodynamic modelling

Thermodynamic calculations were done using the console mode of the ThermoCalc software package [20] version 2015b. The used thermodynamic models are explained as follows. The Gibbs energy G^{α} of a phase α can be expressed as

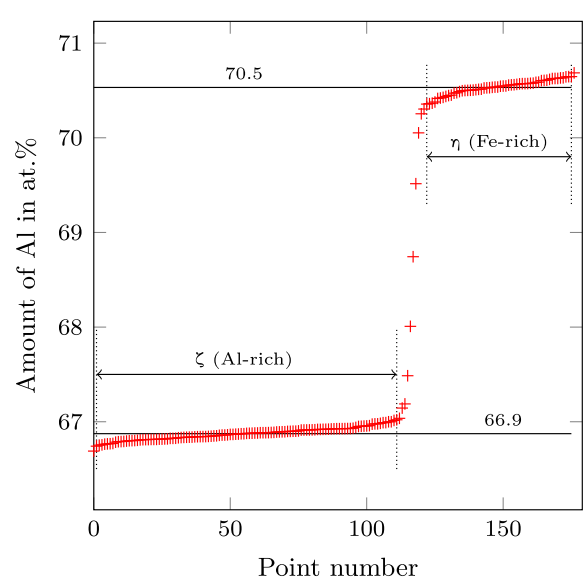

Fig. 2. EPMA/WDS results of the two-phase sample ζ + η of the Al-Fe system (67.0 at.% Al) annealed at a temperature of 1073 K.

$$G^{\alpha} = {}^{\mathrm{srf}}G^{\alpha} - T\cdot{}^{\mathrm{cfg}}S^{\alpha} + {}^{\mathrm{E}}G^{\alpha} + {}^{\mathrm{phys}}G^{\alpha}. \quad (1)$$

${}^{\mathrm{srf}}G^{\alpha}$ is the Gibbs energy of the 'surface of reference' that represents the Gibbs energy of the *end-members* of the phase. The configurational entropy of the phase is expressed by ${}^{\mathrm{cfg}}S^{\alpha}$ and deviation from ideal behaviour of the constituents of the phase is expressed by the excess Gibbs energy ${}^{\mathrm{E}}G^{\alpha}$. Any physical contribution to Gibbs energy, e.g. magnetic transition, is included in the ${}^{\mathrm{phys}}G^{\alpha}$ term. The Gibbs energy of liquid is described using the substitutional model [21]. Using models with mixing of Al and Fe in only one sublattice for the phases liquid, ε, ζ, η and θ, their corresponding Gibbs energy descriptions are equivalent to the substitutional model.

The magnetic contribution to Gibbs energy in A2 is modelled using an empirical approach as described by Hillert and Jarl [22] and by Inden [23]. With that, the magnetic contribution to Gibbs energy ${}^{\mathrm{mag}}G$ is expressed as

$${}^{\mathrm{mag}}G = n\cdot RT\cdot f(\tau)\cdot\ln(\beta+1), \quad (2)$$

where n is number of atoms per formular unit that have the average magnetic moment β [21]. The function $f(\tau)$ is given by

$$f(\tau)=\begin{cases} 1-\frac{1}{A}\cdot\left[\frac{79\tau^{-1}}{140p}+\frac{474}{497}\cdot\left(\frac{1}{p}-1\right)\cdot\left(\frac{\tau^{3}}{6}+\frac{\tau^{9}}{135}+\frac{\tau^{15}}{600}\right)\right], \text{ for } \tau<1 \\ -\frac{1}{A}\cdot\left(\frac{\tau^{-5}}{10}+\frac{\tau^{-15}}{315}+\frac{\tau^{-25}}{1500}\right), \text{ for } \tau\geq 1 \end{cases} \quad (3)$$

with

$$A = \frac{518}{1125}+\frac{11692}{15975}\cdot\left(\frac{1}{p}-1\right), \quad (4)$$

where $\tau = \frac{T}{T_C}$ with the Curie temperature T_C and p is an empirical constant representing the ratio between the contributions of long-range order and short-range order to magnetic enthalpy. Based on the findings of Inden (1981) [23], $p = 0.4$ is used for bcc and $p = 0.28$ is used for fcc materials which is adopted in the present work as well.

2.5. Modelling order/disorder in bcc

The ordering of the disordered A2 phase to B2 phase in the Al-Fe system is modelled using the compound energy formalism [24]. The ordering of $D0_3$ is not included in the present work. The current thermodynamic description is derived as part of a description of the Al-Fe-Mg-Si system, for which the $D0_3$ ordering can not be properly modelled in higher-order systems due to lack of experimental data, i.e. for the Al-Fe-Si system. Therefore, it was decided to only derive thermodynamic parameters for calculation of the A2/B2 ordering with focus on properly reproducing the corresponding transition temperature.

For the B2 model, the A2 sublattice is divided into two sublattices

$$(\mathrm{Al},\mathrm{Fe})_{0.5}(\mathrm{Al},\mathrm{Fe})_{0.5}. \quad (5)$$

The Gibbs energy of A2 and B2 is expressed in one description using the partioning Gibbs energy model as explained by Ansara et al. (1997) [25] and by Dupin and Ansara (1999) [26]. In that approach, the ordering contribution to Gibbs energy is separated from the description of the disordered phase. The Gibbs energy is then given by

$$G = G^{\mathrm{A2}} + \Delta G^{\mathrm{B2}} \quad (6)$$

with the ordering contribution expressed as

$$\Delta G^{\mathrm{B2}} = G^{\mathrm{B2}}(y_i) - G^{\mathrm{B2}}(y_i = x_i), \quad (7)$$

where y_i is the site fraction and x_i is the mole fraction of element i. With such expression, it is ensured that ΔG^{B2} will be zero in the disordered state when it holds $y_i = x_i$.

From crystallography, the B2 phase has two equivalent sites and thus, the Gibbs energy of the two corresponding endmembers must be equal

$${}^{\circ}G^{\mathrm{B2}}_{\mathrm{Al:Fe}} = {}^{\circ}G^{\mathrm{B2}}_{\mathrm{Fe:Al}}. \quad (8)$$

The same argument holds for the parameters of the excess Gibbs energy, for example

$$L^{\mathrm{B2}}_{\mathrm{Al,Fe:*}} = L^{\mathrm{B2}}_{\mathrm{*:Al,Fe}}, \quad (9)$$

where the star symbol indicates the presence of both, Al and Fe, on the respective sublattice.

The Gibbs energy of the other two end-members of the B2 phase, ${}^{\circ}G^{\mathrm{B2}}_{\mathrm{Al:Al}}$ and ${}^{\circ}G^{\mathrm{B2}}_{\mathrm{Fe:Fe}}$, are equal to the Gibbs energies of the A2 end-members ${}^{\circ}G^{\mathrm{A2}}_{\mathrm{Al}}$ and ${}^{\circ}G^{\mathrm{A2}}_{\mathrm{Fe}}$, respectively, and therefore, they are excluded in the B2-phase description.

The chemical ordering can also influence the magnetic properties. The dependence of magnetic ordering on chemical ordering can be included in the thermodynamic description by using

$$T_C = T_C^{\mathrm{A2}} + \Delta T_C^{\mathrm{B2}} \quad (10)$$

and

$$\beta = \beta^{\mathrm{A2}} + \Delta\beta^{\mathrm{B2}} \quad (11)$$

for expressing T_C and β also for the ordered phase. As explained above, symmetrical parameters in both functions must be used in case of the B2 phase, namely for the parameters $T^{\mathrm{B2_BCC}}_{\mathrm{CAl:Fe:Va}}$, ${}^{0}T^{\mathrm{B2_BCC}}_{\mathrm{CAl,Fe:*:Va}}$, $\beta^{\mathrm{B2_BCC}}_{\mathrm{Al:Fe:Va}}$, ${}^{0}\beta^{\mathrm{B2_BCC}}_{\mathrm{Al,Fe:*:Va}}$, ${}^{1}\beta^{\mathrm{B2_BCC}}_{\mathrm{Al,Fe:*:Va}}$ and ${}^{2}\beta^{\mathrm{B2_BCC}}_{\mathrm{Al,Fe:*:Va}}$.

2.5.1. Calculation of the Curie temperature for A2 $\left(T_C^{A2}\right)$ and the transformation temperature for A2/B2 $(T_{A2/B2})$

The Gibbs energy G^{bcc} of the bcc phases A2 $(\mathrm{Al,Fe})_1(\mathrm{Va})_3$ and B2 $(\mathrm{Al,Fe})_{0.5}(\mathrm{Al,Fe})_{0.5}(\mathrm{Va})_3$ is given by equation (6) with

$$\begin{aligned} G^{\mathrm{A2}} &= x_{\mathrm{Al}}\cdot G^{\mathrm{A2}}_{\mathrm{Al:Va}} + (1-x_{\mathrm{Al}})\cdot G^{\mathrm{A2}}_{\mathrm{Fe:Va}} + RT[(x_{\mathrm{Al}})\ln(x_{\mathrm{Al}}) \\ &\quad + (1-x_{\mathrm{Al}})\ln(1-x_{\mathrm{Al}})] + x_{\mathrm{Al}}\cdot(1-x_{\mathrm{Al}})\cdot\Big[{}^{0}L^{\mathrm{A2}}_{\mathrm{Al,Fe:Va}} \\ &\quad + (2x_{\mathrm{Al}}-1)\cdot{}^{1}L^{\mathrm{A2}}_{\mathrm{Al,Fe:Va}}\Big] + G^{\mathrm{mag}}(y'_{\mathrm{Al}} \\ &= x_{\mathrm{Al}}, x_{\mathrm{Al}}) \end{aligned} \quad (12)$$

and

$$\Delta G^{B2} = \left(-4x_{Al}y'_{Al} + 2x_{Al} + 2y'^2_{Al}\right)\cdot G^{B2}_{Al:Fe:Va} + 0.5\cdot RT\cdot\left[(y'_{Al})\ln(y'_{Al}) + (1-y'_{Al})\ln(1-y'_{Al}) + (2x_{Al}-y'_{Al})\ln(2x_{Al}-y'_{Al}) + (1-2x_{Al}+y'_{Al})\ln(1-2x_{Al}+y'_{Al})\right] + \left(-4x^2_{Al} + 4x_{Al}y'_{Al} + 2x_{Al} - 2y'^2_{Al}\right)\cdot {}^0L^{B2}_{Al,Fe:*:Va} + \left(-16x^3_{Al} + 24x^2_{Al}y'_{Al} + 12x^2_{Al} - 12x_{Al}y'^2_{Al} - 12x_{Al}y'_{Al} - 2x_{Al} + 6y'^2_{Al}\right)\cdot {}^1L^{B2}_{Al,Fe:*:Va} + \left(-64x^4_{Al} + 128x^3_{Al}y'_{Al} + 64x^3_{Al} - 96x^2_{Al}y'^2_{Al} - 96x^2_{Al}y'_{Al} - 20x^2_{Al} + 32x_{Al}y'^3_{Al} + 48x_{Al}y'^2_{Al} + 20x_{Al}y'_{Al} + 2x_{Al} - 8y'^4_{Al} - 10y'^2_{Al}\right) + G^{mag}(y'_{Al}, x_{Al}) - \left(-2x^2_{Al} + 2x_{Al}\right)\cdot G^{B2}_{Al:Fe:Va} - RT\cdot\left[(x_{Al})\ln(x_{Al}) + (1-x_{Al})\ln(1-x_{Al})\right] - \left(-2x^2_{Al} + 2x_{Al}\right)\cdot {}^0L^{B2}_{Al,Fe:*:Va} - \left(-4x^3_{Al} + 6x^2_{Al} - 2x_{Al}\right)\cdot {}^1L^{B2}_{Al,Fe:*:Va} - \left(-8x^4_{Al} + 16x^3_{Al} - 10x^2_{Al} + 2x_{Al}\right)\cdot {}^2L^{B2}_{Al,Fe:*:Va} - G^{mag}(y'_{Al} = x_{Al}, x_{Al}), \tag{13}$$

using the relations shown by equation (8),

$$x_{Fe} = 1 - x_{Al}, y'_{Fe} = 1 - y'_{Al}\ ,\ y''_{Al} = 2x_{Al} - y'_{Al} \text{ and } y''_{Fe} = 1 - y''_{Al}, \tag{14}$$

${}^1L^{B2}_{Al,Fe:*:Va} = {}^1L^{B2}_{*:Al,Fe:Va}$ and ${}^2L^{B2}_{Al,Fe:*:Va} = {}^2L^{B2}_{*:Al,Fe:Va}$. Here, the site fractions of the first and second sublattice are marked by y' and y'', respectively.

If the bcc phase is in ordered B2 state, the resulting Gibbs energy will show a minimum at $y'_{Al,1}$ and $y'_{Al,2} = 2x_{Al} - y''_{Al,1}$, and one maximum at $y'_{Al} = x_{Al}$. The respective Gibbs energy curve is shown as an example in Fig. 3a for $x_{Al} = 0.35$ and $T = 1000$ K. Above the transition temperature $T_{A2/B2}$, the Gibbs energy curve will show one minimum at $y'_{Al} = x_{Al}$, which can be seen in Fig. 3b for $T = 1800$ K. The transition temperature $T_{A2/B2}$ is reached when the maximum at $y'_{Al} = x_{Al}$ transforms to a minimum that is the case when

$$\frac{\partial^2 G^{bcc}}{\partial y'^2_{Al}}\Big|_{y'_{Al}=x_{Al}} = 0 \tag{15}$$

is fulfilled, which is illustrated by a plot of $\frac{\partial^2 G^{bcc}}{\partial y'^2_{Al}}$ for the temperatures 1000 K, $T_{A2/B2}$ and 1800 K for the composition $x_{Al} = 0.35$ as shown in Fig. 3c.

The second derivate of G^{bcc} with respect to y'_{Al} is given by

$$\frac{\partial^2 G^{bcc}}{\partial y'^2_{Al}} = 4\cdot G^{B2}_{Al:Fe:Va} + 0.5\cdot RT\cdot\left[\frac{1-2x_{Al}+2y'_{Al}}{y'_{Al} - 2x_{Al}y'_{Al} + y'^2_{Al}} + \frac{2x_{Al}+1-2y'_{Al}}{2x_{Al} - 2x_{Al}y'_{Al} - y'_{Al} + y'^2_{Al}}\right] + (-4)\cdot {}^0L^{B2}_{Al,Fe:*:Va} + (-24x_{Al}+12)\cdot {}^1L^{B2}_{Al,Fe:*:Va} + \left(-192x^2_{Al} + 192x_{Al}y'_{Al} + 96x_{Al} - 96y'^2_{Al} - 20\right)\cdot {}^2L^{B2}_{Al,Fe:*:Va} + \frac{\partial^2 G^{mag}}{\partial y'^2_{Al}} \tag{16}$$

that reduces at $y'_{Al} = x_{Al}$ to

$$\frac{\partial^2 G^{bcc}}{\partial y'^2_{Al}}\Big|_{y'_{Al}=x_{Al}} = 4\cdot G^{B2}_{Al:Fe:Va} + 0.5\cdot RT\cdot\left[\frac{2}{-x^2_{Al} + x_{Al}}\right] + (-4)\cdot {}^0L^{B2}_{Al,Fe:*:Va} + (-24x_{Al}+12)\cdot {}^1L^{B2}_{Al,Fe:*:Va} + \left(-96x^2_{Al} + 96x_{Al} - 20\right)\cdot {}^2L^{B2}_{Al,Fe:*:Va} + \frac{\partial^2 G^{mag}}{\partial y'^2_{Al}}\Big|_{y'_{Al}=x_{Al}}. \tag{17}$$

The second derivative of G^{mag} with respect to y'_{Al} is given by

$$\frac{\partial^2 G^{mag}}{\partial y'^2_{Al}} = RT\cdot\left(\frac{\partial^2 f(\tau)}{\partial y'^2_{Al}}\cdot\ln(\beta+1) + 2\cdot\frac{\partial f(\tau)}{\partial y'_{Al}}\cdot\frac{\partial\ln(\beta+1)}{\partial y'_{Al}} + f(\tau)\cdot\frac{\partial^2\ln(\beta+1)}{\partial y'^2_{Al}}\right). \tag{18}$$

The needed partial derivatives of $f(\tau)$ and $\ln(\beta+1)$ are listed in the appendix in section AppendixA.

In this work, the ordering contribution to Gibbs energy is modelled by using the parameters $G^{B2}_{Al:Fe:Va}$, ${}^1L^{B2}_{Al,Fe:*:Va}$, ${}^2L^{B2}_{Al,Fe:*:Va}$, $T^{B2}_{CAl:Fe} = T^{B2}_{CFe:Al}$ and ${}^0T^{B2}_{CAl,Fe:*} = {}^0T^{B2}_{C*:Al,Fe}$. The parameters for β^{B2} were adopted from Sundman et al. [14].

By examining equation (17), it can be seen that ${}^0L^{B2}_{Al,Fe:*:Va}$ has no composition dependent contribution to $\frac{\partial^2 G^{bcc}}{\partial y'^2_{Al}}$. Therefore, it can not be used to describe the composition dependency of the A2/B2 transition temperature. In agreement with the modified sublattice formalism [26], the ${}^0L^{B2}_{Al,Fe:*:Va}$ parameter was set to zero in this work.

The calculation of the two transition temperatures T^{A2}_C and $T_{A2/B2}$ was done in the following way. First, the Curie temperature of the disordered phase T^{A2}_C was obtained by fitting the parameters ${}^0T^{A2}_{CAl,Fe}$, ${}^1T^{A2}_{CAl,Fe}$ and ${}^2T^{A2}_{CAl,Fe}$ of the expression

$$T^{A2}_C = (1-x_{Al})\cdot T^{A2}_{CFe} + \left(x_{Al} - x^2_{Al}\right)\cdot {}^0T^{A2}_{CAl,Fe} + \left(-2x^3_{Al} + 3x^2_{Al} - x_{Al}\right)\cdot {}^1T^{A2}_{CAl,Fe} + \left(-4x^4_{Al} + 8x^3_{Al} - 5x^2_{Al} + x_{Al}\right)\cdot {}^2T^{A2}_{CAl,Fe} \tag{19}$$

to experimental data from Stein and Palm [27]. In a second step, equation (15) was used to calculate $T_{A2/B2}$ and to optimise the mentioned ordering parameters using also the experimental data from Ref. [27]. The calculated Curie temperatures and A2/B2 temperatures are shown in Fig. 4 for $0 \le x_{Al} \le 0.5$ in comparison to the calculated trend of $T_{A2/B2}$ using the thermodynamic parameters reported by Seierstein [28], where no explicit ordering parameter was used. The $T_{A2/B2}$ transition temperature reported in Ref. [14] is also shown in Fig. 4. Both calculated temperature trends of this work are in perfect agreement with reported experiments. The derived thermodynamic parameters are listed in Table 4.

2.6. Thermodynamic modelling of intermetallics

The c_p description of the intermetallic phases ζ, η and θ is based on the measured heat capacity values reported elsewhere [8,12].

2.6.1. Modelling of ζ

The structure of $FeAl_2$ was investigated several times [4,29,30]. Based on the latest structure analysis [30], the thermodynamic model

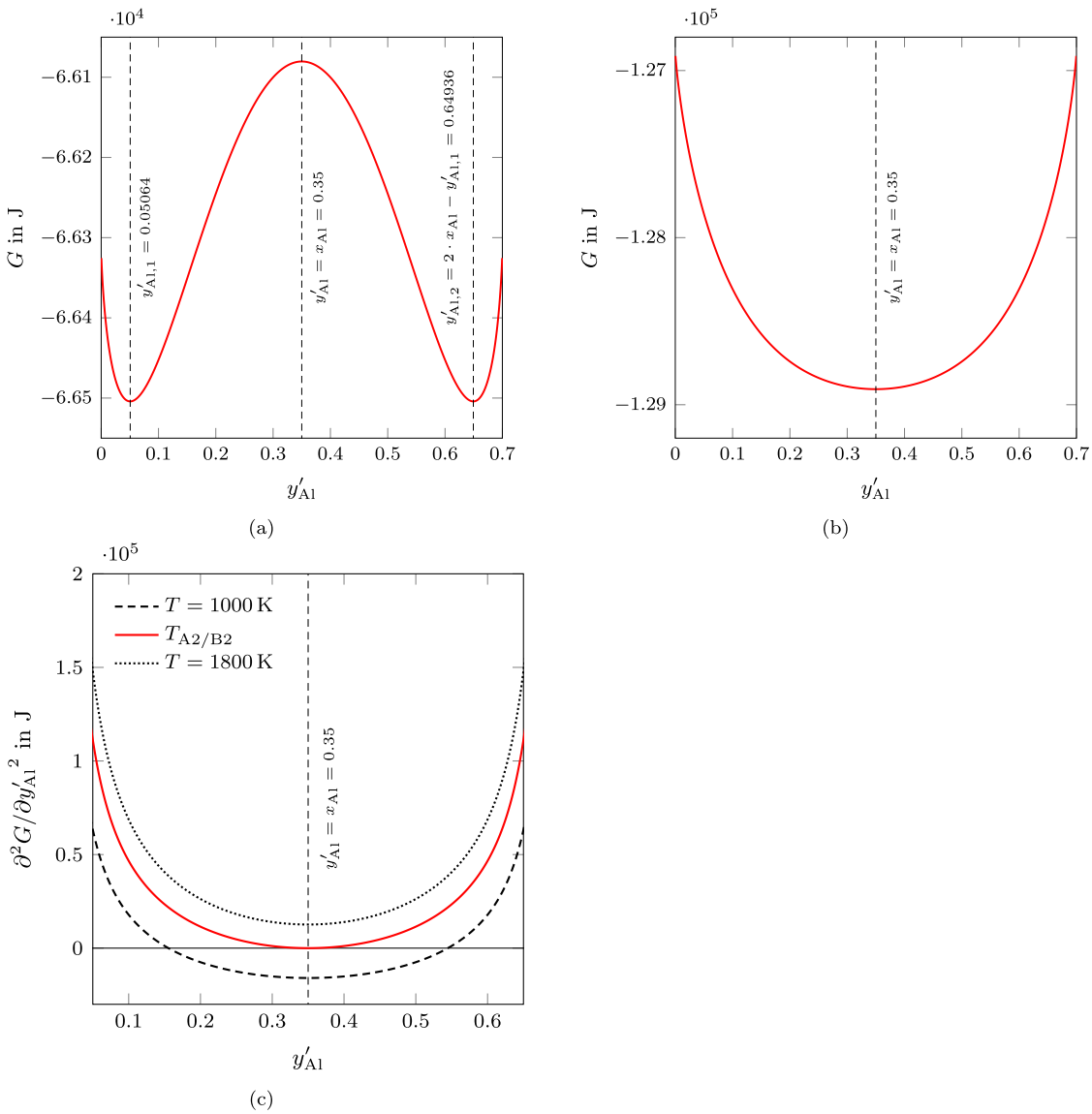

Fig. 3. Calculated thermodynamic functions of B2_BCC at $x_{Al} = 0.35$. a) G at $T = 1000$ K, b) G at $T = 1800$ K, and c) $\partial^2 G/\partial y'^2_{Al}$.

$$(Al)_{12}(Fe)_5(Al, Fe)_2 \tag{20}$$

was chosen, which gives a composition range of ζ between approximately 63.2 and 73.7 at.% Al. The existence of thermal vacancies in ζ, η and θ was discussed previously [7]. It was suggested that the Gibbs energy of the Fe-rich compositions are lowered in comparison to the Al-rich ones because of the formation of thermal vacancies at Al-sites of the structure, which is the reason for the inclining of the homogeneity ranges towards Fe-rich compositions with increasing temperature.

Thermal vacancies could be modelled using the approach proposed by Franke [31] or by Guan and Liu [32]. However, there is no proof of the existence of thermal vacancies and the knowledge about their amount in the intermetallic phases ζ, η and θ. Therefore, it was decided to use a simpler solution to describe the measured heat capacity of these phases.

In principle, the contribution of thermal vacancy formation to c_p can be expressed using an exponential equation (see Ref. [12] for details). However, as it is not possible to directly include such formula into the thermodynamic description using current thermodynamic software, the measured heat capacity was fitted by

$$c_p = -c - 2d \cdot T - 6f \cdot T^2 - 12e \cdot T^3 \tag{21}$$

for $T \geq 298.15$ K. Using this formula, it can be seen that the heat capacity increases strongly with increasing temperature and thus lowers the Gibbs energy of an end-member in comparison to an end-member without a non-linear increase of c_p. Therefore, a composition dependent non-linear increase of c_p can be used to model the shifting of the homogeneity ranges based on the measured thermodynamic function.

In case of ζ, the obtained c_p equation was used for modelling c_p of the Fe-rich endmember $Al_{12}Fe_5Fe_2$. The heat capacity of the Al-rich endmember $Al_{12}Fe_5Al_2$ increases linearly with temperature in case the obtained c_p equation without thermal vacancy contribution is used (see [12, Table 1]). In that way, c_p will increase non-linearly in Fe-rich compositions at higher temperatures. This approach suggests a composition dependent non-linear increase of the heat capacity. However, the heat capacity was measured only

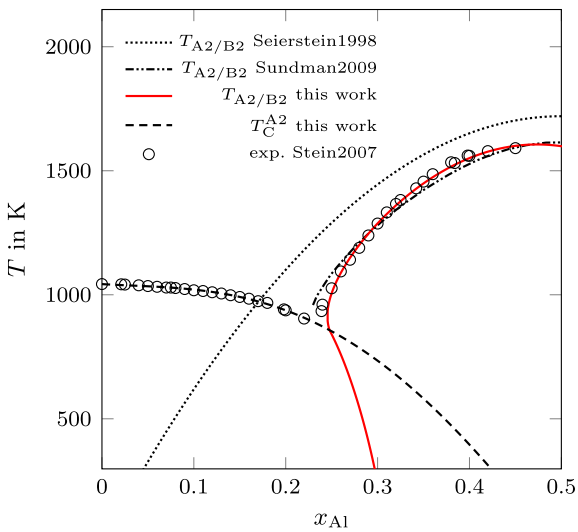

Fig. 4. Calculated trend of T_C^{A2} and $T_{A2/B2}$ up to 50 at.% Al in comparison to experimental data from Stein and Palm [27] and the calculated trend of $T_{A2/B2}$ using the thermodynamic parameters reported by Seierstein [28] and by Sundman et al. [14].

for one composition, which is used here to describe the heat capacity of the Fe-rich compositions. Therefore, the calculated heat capacity for the sample composition would be lower than the measured one as it is the result of a summation of the c_p function from both end-members.

One could also use a stronger increasing c_p for the Fe-rich end-member to obtain the correct heat capacity for the sample composition. However then, due to the exponentially increase of c_p with temperature, it would stabilise the Fe-rich end-member too strongly at high temperature. Therefore, a compromise was made here to reproduce both - the lowering of Gibbs energy of Fe-rich compositions and the measured experimental heat capacity - in a good manner. The obtained heat capacities of the endmembers of ζ are shown in Fig. 5a.

2.6.2. Modelling of η

A detailed structure investigations were done on a sample with the nominal composition $Al_{71.5}Fe_{28.5}$ by Burkhardt et al. [33]. The previously found space group *Cmcm* [34–36] was confirmed in this study and one additional atomic position was found for Al. The corresponding thermodynamic model would be

$$(Al)_8(Fe)_4(Al,Va)_4(Al,Va)_8 \tag{22}$$

with possible constitutional vacancies in two sublattices not considering a new work [11] proposing mixed channel occupation by Fe and Al with an approximately concentration independent atomic density. However, not enough information on the occupancy of the vacancy positions are available. Therefore, site occupancy on two sublattices can not be modelled accurately and the thermodynamic model must be simplified. For the thermodynamic assessment, mixing of Al and Va on one sublattice was used resulting in the model

$$(Al)_8(Fe)_4(Al,Va)_{12}, \tag{23}$$

which gives a composition range between approximately 66.6 and 83.3 at.% Al. One problem occurred during the assessment of the Al-Fe system. It was not possible to obtain a good reproduction of the η-phase homogeneity range and an equilibrium between θ and the Al-rich FCC phase at the same time. Instead, the Al-rich endmember of the η-model was always too stable resulting in an equilibrium of η (Al-rich) + FCC. Therefore, the maximum composition range of the η-phase model was narrowed assuming maximal concentration of Al to be 75 at.% Al by using the model

$$(Al)_8(Fe)_4(Al,Va)_4. \tag{24}$$

The heat capacity of η was modelled similar to that of ζ. Here, the thermodynamic model includes the presence of constitutional vacancies in one sublattice to model the observed homogeneity range of the phase. As thermal vacancies are not included in the thermodynamic description, the non-linear increase of c_p with increasing amount of thermal vacancies is approximated by setting the c_p of the Fe-rich endmember $Al_8Fe_4Va_4$ equal to a fit of equation (21) to the measured c_p values for the same reasons as discussed in case of ζ. The c_p of the Al-rich endmember $Al_8Fe_4Al_4$ is linear increasing with temperature as explained above for ζ. The obtained heat capacities for both endmembers are shown in Fig. 5b. It should be mentioned that the observed phase transitions in η at temperatures around 600 K (see Ref. [8] for details) were not taken into account in the thermodynamic description of this work. Only tentative informations are available for the low temperature part of the homogeneity range of η [9–11]. Therefore, the order/disorder transitions in η could be modelled in a future assessment.

Table 1
Temperatures of extrapolated onset points (OP) at a heating rate of 0 K/min rounded to the next nearest unit. Invariant reactions are marked with an asterisk (*).

proposed Reaction	comp. in at.% Al	OP	Temperature in K
L + B2 → L	56.0	III	1551
B2 + ε → L + B2	56.0	II	1507
*L + B2 →ε			1507
L + ε → L	61.6	III	1498
	63.0	IV	1479
	65.9	1-IV	1457
		2-IV	1457
	66.5	IV	1445
	68.0	III	1442
ε→ L + ε	61.6	II	1472
	63.0	III	1443
ε + η → L + ε	66.5	III	1431
	67.0	II	1432
	65.9	1-III	1430
		2-III	1429
	68.0	II	1430
*L →ε + η			1430
η→ L + η	70.0	I	1430
η + θ → L + η	71.7	I	1426
	72.7	I	1425
	73.7	I	1425
*L + η→θ			1425
ε + ζ → ε + η	65.9	1-II	1425
		2-II	1424
ζ + η → ε + η	66.5	II	1421
	67.0	I	1421
	68.0	I	1420
*ε + η → ζ			1422
ζ→ζ + η	66.5	I	1421
ε + ζ → ε	63.0	II	1412
B2 + ζ → B2 + ε	56.0	I	1369
B2 + ζ → ε + ζ	61.6	I	1369
	63.0	I	1370
	65.9	1-I	1369
		2-I	1369
*ε→ B2 + ζ			1369
θ→ L + θ	75.3	I	1406

2.6.3. Modelling of θ

According to crystallography [37,38], the thermodynamic model of θ would be

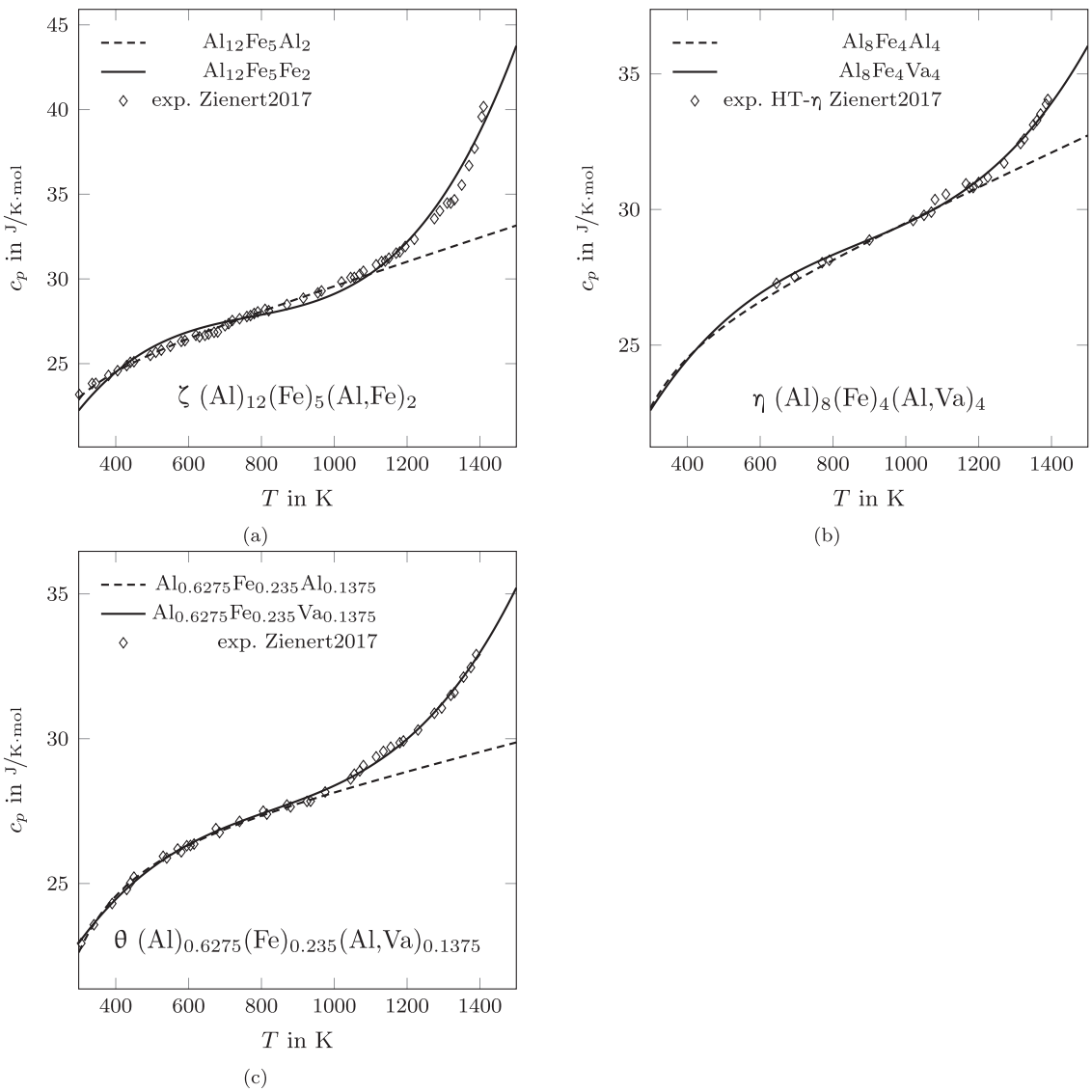

Fig. 5. Molar heat capacity of the endmembers of the modelled intermetallic phases a) ζ, b) η and c) θ in comparison to the experimental values reported by Zienert et al. (2017) [12].

$$(\mathrm{Al})_{74}(\mathrm{Fe})_{24}(\mathrm{Al},\mathrm{Va})_4 \tag{25}$$

describing the homogeneity range between approximately 75.5 and 76.5 at.% Al. In this work, the minimal determined Al-content of θ is ≈74 at.% Al (see section 3.1.2) in agreement with literature [7]. Therefore, the model

$$(\mathrm{Al})_{0.6275}(\mathrm{Fe})_{0.235}(\mathrm{Al},\mathrm{Va})_{0.1375} \tag{26}$$

proposed by Seierstein [28] with a homogeneity range between approximately 72.8 and 76.5 at.% Al was accepted. The same model approach as in case of η was used to describe heat capacity which is shown for both endmembers in Fig. 5c.

3. Results and discussion

3.1. Experimental phase diagram

3.1.1. Liquidus, solidus and solvus temperatures

Liquidus, solidus and solvus temperatures were determined by DTA experiments for samples between 56.0 at.% Al and 75.3 at.% Al. DTA curves obtained using a heating rate of 2 K/min for sample compositions between 56.0 at.% Al and 65.9 at.% Al are shown in Fig. 6 and for sample compositions between 66.5 at.% Al and 75.4 at.% Al in Fig. 7. Extrapolated onset point temperatures to a heating rate of 0 K/min are listed in Table 1. If possible, the temperatures of the invariant reactions were calculated as mean values of extrapolated onset points from different samples. In such cases, the standard deviation σ was calculated for each invariant temperature. Both values are listed in Table 1. The standard deviation was in the range between 0.46 K $\leq \sigma \leq$ 0.9 K. Therefore, the expected minimal accuracy of the given temperatures is ±1 K. Only in the case of the invariant reaction $\varepsilon + \eta \rightarrow \zeta$, the standard deviation was 2.16 K. For the 65.9 at.% Al sample, the temperature of the determined onset points was approximately 3–5 K higher then the ones for the samples between 66.5 at.% Al and 68.0 at.% Al. It should be mentioned that the lower temperatures are in a very good agreement with the reported temperature of the invariant reaction by Li at al. (2016) [6] and Han et al. (2016) [7]. However, a slightly higher value is recommended in the present work because it is an

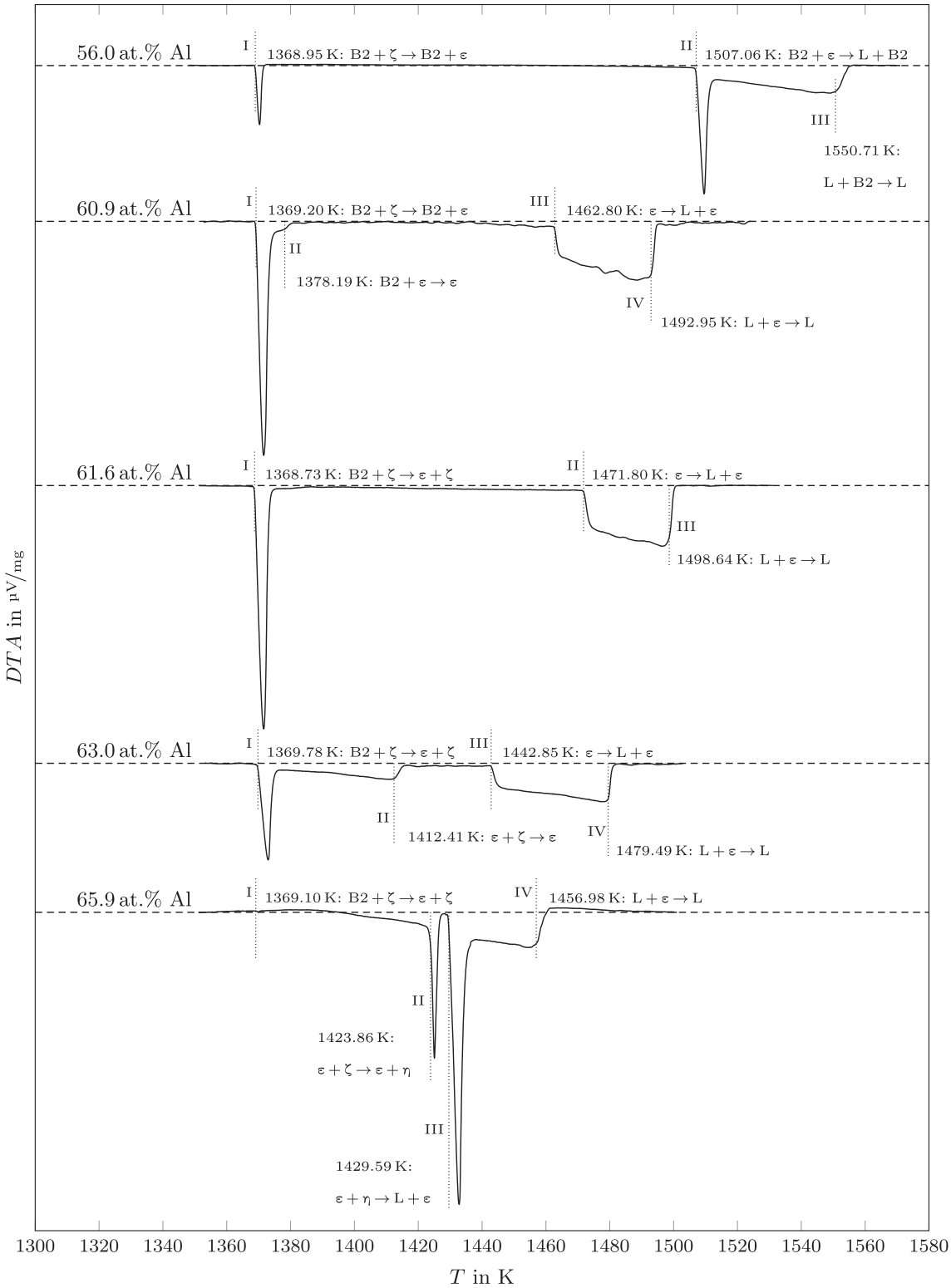

Fig. 6. Measured DTA signals with a heating rate of 2 K/min and determined onset points of the samples between 56.0 at.% Al and 65.9 at.% Al.

average value obtained for several compositions and it is in perfect agreement with the second measurement of another 65.9 at.% Al sample.

A more detailed comparison between experimental values of this work and from literature will be given together with the results of the thermodynamic assessment of the Al-Fe system in section 3.3.

The character of the θ-phase formation was stated as unknown in the evaluation by Kattner and Burton (1993) [3]. No decision could be made whether θ melts congruently or peritecticaly according to L + η → θ. In subsequent experimental works [6,7], the peritectic formation was preferred, which is also recommended in

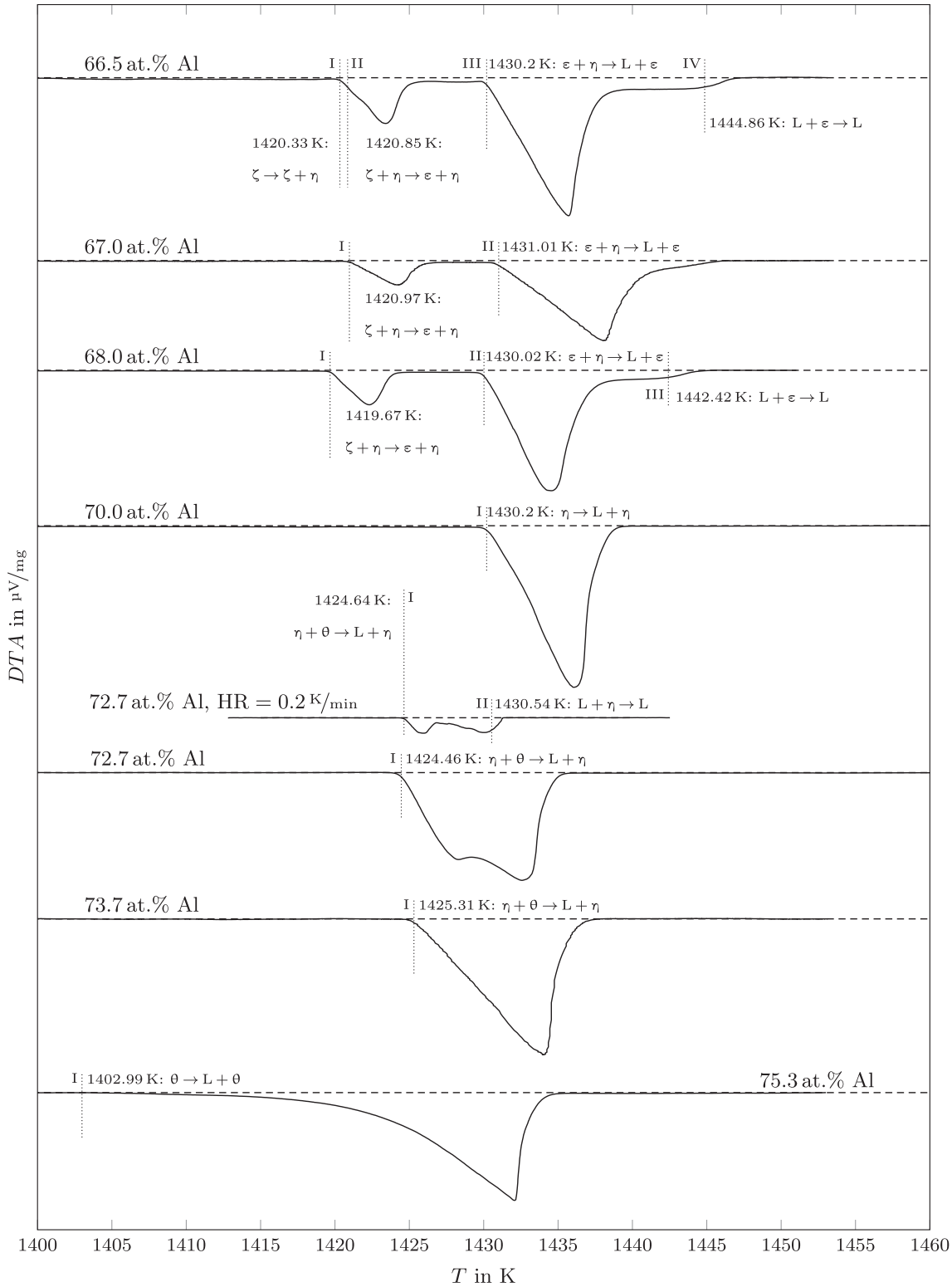

Fig. 7. Measured DTA signals and determined onset points of the samples between 66.5 at.% Al and 75.3 at.% Al. If not otherwise declared, the heating rate (HR) was always 2 K/min.

this work. In Fig. 8, SEM micrographs of the 73.7 at.% Al sample after DTA are shown. Small grains of η (see EBSD map in Fig. 8b) are embedded in a matrix of large uniform θ-phase grains. No areas of mixed grains of η + θ were found, which would be a hint for a eutectic L → η + θ transition.

3.1.2. Homogeneity ranges

The homogeneity ranges of the intermetallic phases ζ, η and θ, and to some extend of B2 and ε were determined by composition measurements using EDS and WDS on quenched samples from a temperature of 1073 K up to near the respective melting points as described in section 2.2; the results are listed in Table 2. As we

observed very similar SEM microstructures in comparison to the reported ones in Ref. [7], we will not discuss them here as they will not bring any new insights related to the phase evolution of the Al-Fe system.

3.1.3. Phase diagram

The experimentally determined phase diagram is presented in Fig. 9. The shifting of the homogeneity ranges to more Fe-rich compositions of ζ, η and θ with increasing temperature [6,7] is confirmed by the present composition measurements. It should be mentioned that the determined composition ranges are closer to the reported values of Han et al. [7] than to the ones of Li et al. [6].

3.2. The high temperature phase ε

3.2.1. Heat capacity of ε

The heat capacity of ε was measured between the temperatures 1395 K and 1455 K. A non-linear increase in c_p was not observed for ε. The determined values are shown in Fig. 10 and listed in Table 3.

3.2.2. Modelling of ε

The structure of ε was determined to be of Hume-Rothery Cu_5Zn_8-type (SG $I\bar{4}3m$, No. 217) [39,40]. Iron excess in a slightly non-stoichiometric Fe_5Al_8 was found on the 24 g positions only [40]. Aluminium excess in Fe_5Al_8 was not investigated so far. The corresponding sublattice model of the Cu_5Zn_8-type for the Al-Fe system is

$$(Fe)_8(Al)_8(Al,Fe)_{12}(Al,Fe)_{24} \tag{27}$$

with mixing of Al and Fe in two sublattices corresponding to the 12e and 24 g Wyckoff positions. From available experimental data on the occupancy of excess atoms in the structure [40], it can not be said if the site occupancies are the same as in the ε-phase in the Cu-Zn system, where the excess atoms of Cu and Zn are not dissolved randomly on 12e and 24 g but in a defined way in one of the two sublattices [41,42].

It was therefore decided to use a simplified thermodynamic model to describe the ε phase. The two sublattices representing the 12e and 24 g sites were combined into one by using the model

$$(Al)_8(Fe)_8(Al,Fe)_{36}. \tag{28}$$

with it, it was possible to reproduce the experimental results on the homogeneity range in a good manner.

The heat capacity description of ε is based on the experimental values and the estimated c_p using Neumann-Kopp calculation for the stoichiometric composition (Fe_5Al_8), which is close to the measured composition, as follows. The calculated heat capacity at $T = 1450$ K is $\approx 10\,J/_{K\cdot mol}$ below the measured one. For the measured temperature range, the heat capacity of ε is linear increasing. If the c_p is extrapolated from the measured values to room temperature, an unrealistic low value of heat capacity would be obtained. Therefore, the c_p of ε is described by a straight line connecting the estimated values near room temperature with the measured values at high temperatures. With that, realistic values of c_p are obtained for the metastable temperature range and the modelled c_p is close to the measured one when ε is stable. The resulting heat capacity of both endmembers is the same as it is shown in Fig. 11.

Table 2
Measured chemical compositions in the Al-Fe system. If not otherwise marked, the compositions were determined by WDS measurements. The former ε-phase is marked as [B2 + ζ].

Phase field	T in K	sample composition in at.% Al	composition in at.% Al	
			phase 1×	phase 2
B2 + [B2 + ζ]	1426	56.0	51.8 (EDS)	58.2 (EDS)
[B2 + ζ] + ζ	1435	63.0	58.6 (EDS)	63.4 (EDS)
	1393	63.0	60.1	64.8
	1273	63.0		65.3
	1173	63.0		65.4
ζ +η	1418	66.5	65.7	
		67.0	66.0	68.5
	1393	66.5	66.4	69.0
		67.0	66.3	69.0
	1273	66.5	66.2	
		67.0	66.5	69.5
	1173	66.5	66.6	69.7
		67.0	66.4	69.7
	1073	67.0	66.9	70.5
η + θ	1418	73.7	71.9	73.9
	1393	73.7	71.5	74.0
		≈72.5	71.7	74.2
	1273	73.7	72.6	74.9
	1173	73.7	72.6	75.4
		75.3	72.2	75.5
	1073	73.7	72.9	75.8

3.3. The calculated phase diagram

The assessment of the Fe-Al system is based on the experimental results reported by Stein and Palm [27], Li et al. [6], Han et al. [7] and on results of this work. Parameters for the phases ε, ζ, η, θ, BCC_B2 and liquid were optimised. The thermodynamic description of A2_BCC and FCC_A1 was accepted from the work of

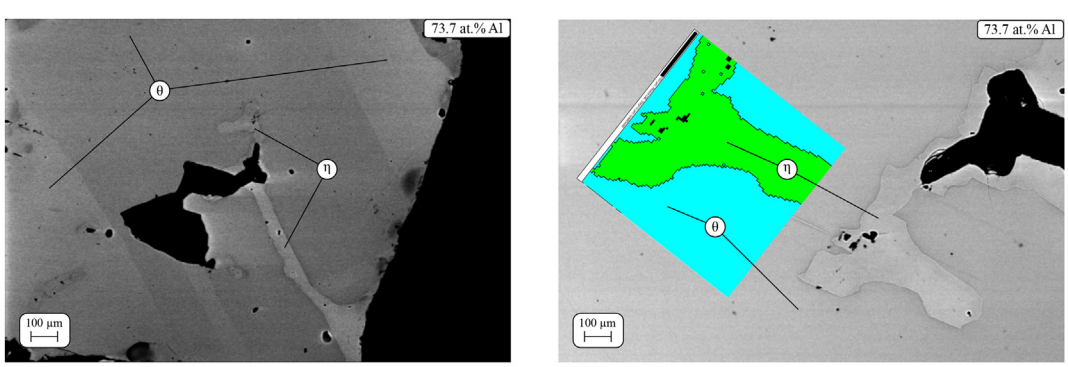

Fig. 8. SEM micrographs of the 73.7 at.% Al sample after DTA showing a) small primary η-phase grains in a matrix of large uniform θ-phase grains and b) the results of an EBSD-mapping (phase contrast) confirming the phase assignment.

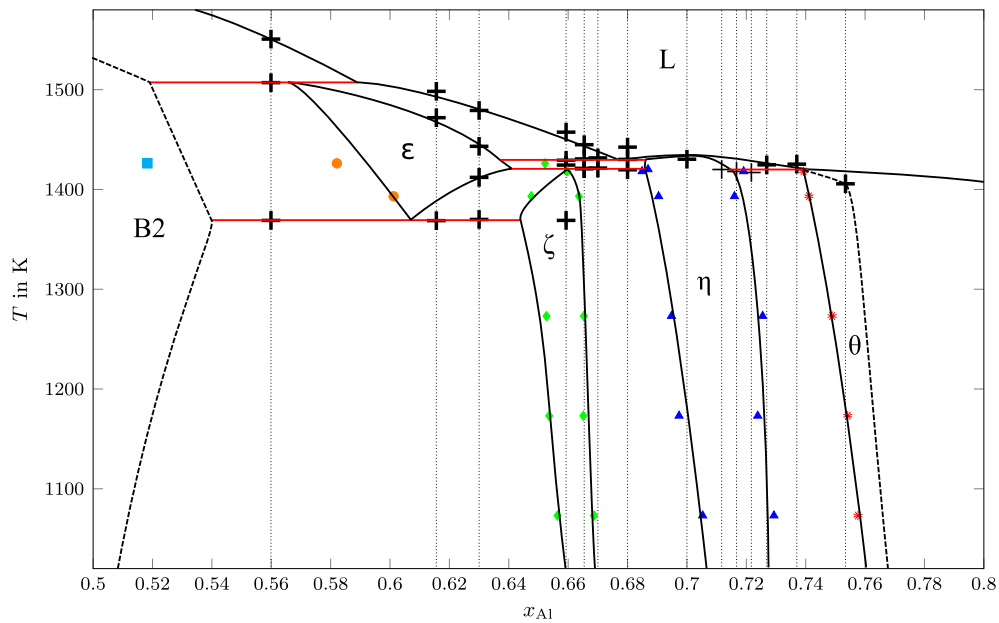

Fig. 9. Experimentally determined Al-rich side of the Al-Fe phase diagram. The crosses mark DTA results whereas the other symbols representing determined compositions measured with EDS and WDS. Investigated sample compositions are marked by dotted lines and the dashed lines showing tentative the end of the homogeneity ranges of B2 and θ, which where not investigated in this study. Compositions not determined in the present work were adopted from Han et al. [7].

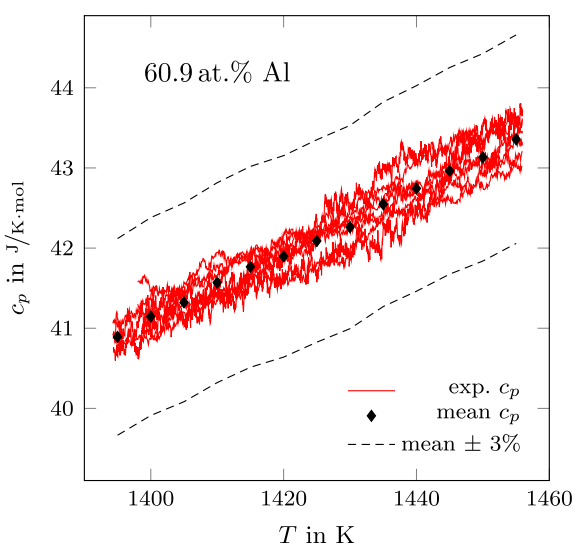

Fig. 10. Measured c_p values of the ε-phase sample (60.9 at.% Al) together with the borders of 3% of the mean value.

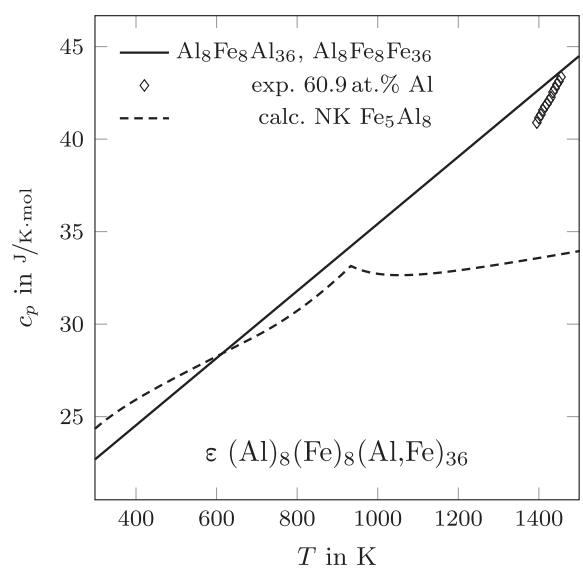

Fig. 11. Molar heat capacity of the endmembers of modelled ε in comparison to the experimental values and calculated heat capacity using Neumann-Kopp (NK) for the composition Fe_5Al_8.

Table 3
Measured c_p of the ε-phase sample (60.9 at.% Al) with T in K and c_p in J/K·mol.

T	c_p	T	c_p	T	c_p	T	c_p	T	c_p
1395	40.9	1410	41.6	1425	42.1	1440	42.7	1455	43.4
1400	41.2	1415	41.8	1430	42.3	1445	43.0		
1405	41.3	1420	41.9	1435	42.6	1450	43.1		

Seierstein [28], only the Curie temperature of A2_BCC was changed as discussed in section 2.5.1. The obtained thermodynamic parameters are listed in Table 4 and the calculated phase diagram of the Fe-Al system is shown in Fig. 12.

The calculated phase diagram between 50 and 80 at.% Al is shown in Fig. 14a in comparison with the experimental results of this work. The transition temperatures determined by DTA are very well reproduced. The three intermetallics ζ, η and ζ were modelled as solution phases. All three phases follow the trend of the experimental determined homogeneity ranges. However, it was not possible to reproduce the shifting to Fe-rich compositions at higher temperatures for ζ and η. The calculated homogeneity range of η is almost constant to temperatures near the melting point not reproducing the experimentally determined broadening above 1000 K. The homogeneity range of ε is very well reproduced by the

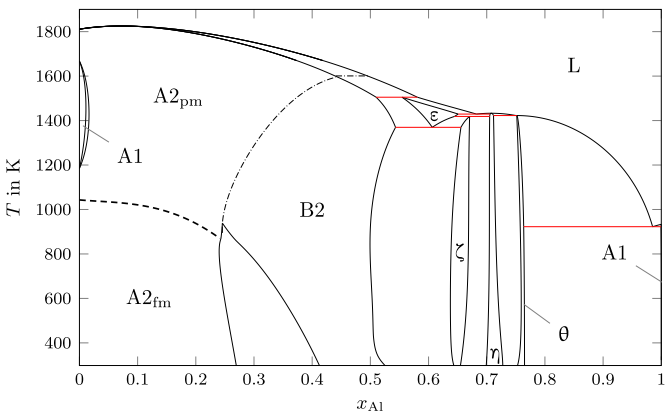

Fig. 12. Calculated phase diagram of the Fe-Al system.

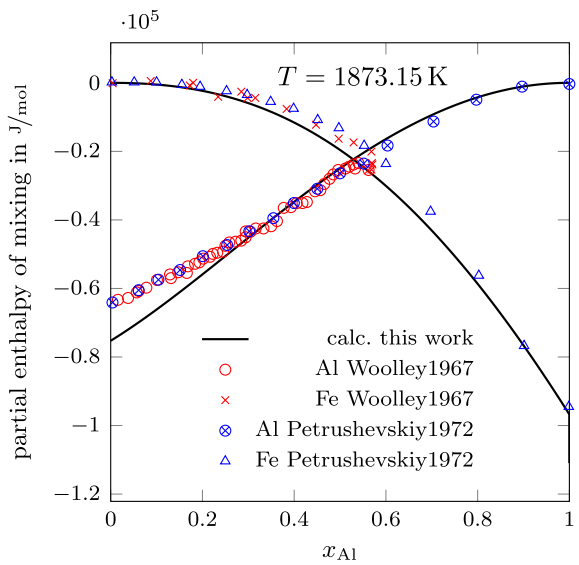

Fig. 13. Calculated partial enthalpy of mixing of Al and Fe in liquid at $T = 1873.15$ K in comparison to experimental values [43,44].

calculation and the overall agreement of the homogeneity ranges of ζ, η and θ in comparison to experimental data is very good as it can be seen in Fig. 14b. The increased solubility of aluminium in B2 is reproduced and its evolution with temperature is in very good agreement with experiments.

According to the calculation, ε and θ form peritectically from melt, η melts congruently and ζ forms in a eutectoid transition, which is in agreement with the reported experimental findings. The calculated temperatures of the invariant transitions are within ±5 K of the reported ones. In Fig. 14c, the corresponding section of the phase diagram is shown in comparison to experimental and calculated invariant temperatures. Comparison of calculated invariant temperatures from the present work and from literature is given in Table 5. It should be mentioned that the calculated temperature difference between the transition temperature of L → ε + η and ε + η → ζ is much better reproduced in this work in comparison to the results reported by Sundman et al. [14].

The trend of the liquidus on the Al-rich side of the phase diagram is in very good agreement with experimental values which can be seen in Fig. 14d. On the Fe-rich side of the Al-Fe phase diagram, the calculated trend of liquidus and solidus temperatures is in very good agreement with experiments which is shown in Fig. 15a. The calculated partial enthalpy of mixing of Al and Fe in liquid at a temperature of 1873 K is shown in Fig. 13 in comparison to experimental values [43,44]. The experimental values are very well reproduced by the calculated ones. Only the partial enthalpy of mixing of Al in Fe-rich liquid is slightly too negative in comparison to the reported values.

At temperatures slightly above the intersection of the Curie temperature of A2 T_C^{A2} with $T_{A2/B2}$, the second-order phase boundary between disordered A2 and ordered B2 transforms into first-order A2 + B2 phase field at the tricritical point (939 K), see Refs. [45–47] for details, which is reproduced by the thermodynamic calculations of the present work. The corresponding part of the phase diagram is shown in Fig. 15b. The upper and lower limit of the A2 + B2 two-phase field (see discussion in Ref. [47]) is marked in Fig. 15b by dotted lines. Below a temperature of 827 K, $D0_3$ should be stable. As the two-sublattice model used in the present thermodynamic description does not support this crystallographic derivate of the parent A2 phase, the A2 + B2 phase field is

Table 4
Thermodynamic parameters of the Al-Fe system.

phase	parameter	value	reference
liquid	$G_{Al}^{liquid} - H_{Al}^{SER}$	GALLIQ	[19]
	$G_{Fe}^{liquid} - H_{Fe}^{SER}$	GFELIQ	[19]
	${}^{0}L_{Al,Fe}^{liquid}$	$-89119.1 + 20.4808 \cdot T$	this work
	${}^{1}L_{Al,Fe}^{liquid}$	$-10752.5 + 8.8022 \cdot T$	this work
	${}^{2}L_{Al,Fe}^{liquid}$	$3103.2 + 0.3221 \cdot T$	this work
epsilon	$G_{Al:Fe:Fe}^{epsilon} - 8H_{Al}^{SER} - 44H_{Fe}^{SER}$	$-1110000 + 5007.8681 \cdot T - 898.4144 \cdot T \cdot \ln(T) - 0.45359288 \cdot T^2$	this work
	$G_{Al:Fe:Al}^{epsilon} - 44H_{Al}^{SER} - 8H_{Fe}^{SER}$	$-276722.3 + 4220 \cdot T - 898.4144 \cdot T \cdot \ln(T) - 0.45359288 \cdot T^2$	this work
	${}^{0}L_{Al:Fe:Al,Fe}^{epsilon}$	$-1838972.4 + 2.5805 \cdot T$	this work
	${}^{1}L_{Al:Fe:Al,Fe}^{epsilon}$	$-4691839 + 3558.4270 \cdot T$	
zeta	$G_{Al:Fe:Al}^{zeta} - 14H_{Al}^{SER} - 5H_{Fe}^{SER}$	$-575167.8 + 2402.6807 \cdot T - 432.1056 \cdot T \cdot \ln(T) - 0.066401105 \cdot T^2 + 1566274.5 \cdot T^{-1}$	this work
	$G_{Al:Fe:Fe}^{zeta} - 12H_{Al}^{SER} - 7H_{Fe}^{SER}$	$-554383 + 893.2723 \cdot T - 173.61174 \cdot T \cdot \ln(T) - 0.6089367 \cdot T^2 + 2.459436E-04 \cdot T^3 - 5.311E-08 \cdot T^4$	this work
eta	$G_{Al:Fe:Al}^{eta} - 12H_{Al}^{SER} - 4H_{Fe}^{SER}$	$-489589.6 + 2134.1820 \cdot T - 377.0176 \cdot T\ln(T) - 0.0494676 \cdot T^2 + 1940352 \cdot T^{-1}$	this work
	$G_{Al:Fe:Va}^{eta} - 8H_{Al}^{SER} - 4H_{Fe}^{SER}$	$-352407.4 + 870.2313 \cdot T - 162.4812 \cdot T \cdot \ln(T) - 0.2465892 \cdot T^2 + 8.1354E-05 \cdot T^3 - 1.551672E-08 \cdot T^4$	this work

Table 4 (*continued*)

phase	parameter	value	reference
theta	$G^{\mathrm{theta}}_{\mathrm{Al:Fe:Al}}$	$-30543.6 + 143.6343 \cdot T$	this work
	$-0.765H^{\mathrm{SER}}_{\mathrm{Al}} - 0.235H^{\mathrm{SER}}_{\mathrm{Fe}}$	$-25.3674 \cdot T\ln(T) - 0.001549605 \cdot T^2 + 163227.5 \cdot T^{-1}$	
	$G^{\mathrm{theta}}_{\mathrm{Al:Fe:Va}}$	$-26391.8 + 90.0757 \cdot T$	this work
	$-0.6275H^{\mathrm{SER}}_{\mathrm{Al}} - 0.235H^{\mathrm{SER}}_{\mathrm{Fe}}$	$-15.2734 \cdot T\ln(T) - 0.0178165 \cdot T^2 + 6.29955\mathrm{E}-06 \cdot T^3 - 1.27236\mathrm{E}-09 \cdot T^4$	
FCC_A1	$G^{\mathrm{FCC_A1}}_{\mathrm{Al:Va}} - H^{\mathrm{SER}}_{\mathrm{Al}}$	GHSERAL	[19]
	$G^{\mathrm{FCC_A1}}_{\mathrm{Fe:Va}} - H^{\mathrm{SER}}_{\mathrm{Fe}}$	GFEFCC	[19]
	${}^{0}G^{\mathrm{FCC_A1}}_{\mathrm{Al,Fe:Va}}$	$-76066.1 + 18.6758 \cdot T$	[28]
	${}^{1}G^{\mathrm{FCC_A1}}_{\mathrm{Al,Fe:Va}}$	$21167.4 + 1.3398 \cdot T$	[28]
	$T^{\mathrm{FCC_A1}}_{\mathrm{CFe:Va}}$	-201	[28]
	$\beta^{\mathrm{FCC_A1}}_{\mathrm{Fe:Va}}$	-2.1	[28]
BCC_A2	$G^{\mathrm{BCC_A2}}_{\mathrm{Al:Va}} - H^{\mathrm{SER}}_{\mathrm{Al}}$	GALBCC	[19]
	$G^{\mathrm{BCC_A2}}_{\mathrm{Fe:Va}} - H^{\mathrm{SER}}_{\mathrm{Fe}}$	GHSERFE	[19]
	$T^{\mathrm{BCC_A2}}_{\mathrm{CFe:Va}}$	1043	[19]
	$\beta^{\mathrm{BCC_A2}}_{\mathrm{Fe:Va}}$	2.22	[19]
	${}^{0}L^{\mathrm{BCC_A2}}_{\mathrm{Al,Fe:Va}}$	$4 \cdot (-30740 + 7.9972 \cdot T)$	[28]
	${}^{1}L^{\mathrm{BCC_A2}}_{\mathrm{Al,Fe:Va}}$	$8 \cdot (368.15)$	[28]
	${}^{0}T^{\mathrm{BCC_A2}}_{\mathrm{CAl,Fe:Va}}$	-2292.17	this work
	${}^{1}T^{\mathrm{BCC_A2}}_{\mathrm{CAl,Fe:Va}}$	-7553.94	this work
	${}^{2}T^{\mathrm{BCC_A2}}_{\mathrm{CAl,Fe:Va}}$	-4435.63	this work
B2_BCC	$G^{\mathrm{B2_BCC}}_{\mathrm{Fe:Al:Va}} - 0.5H^{\mathrm{SER}}_{\mathrm{Al}} - 0.5H^{\mathrm{SER}}_{\mathrm{Fe}}$	-12072.2	this work
	$G^{\mathrm{B2_BCC}}_{\mathrm{Al:Fe:Va}} - 0.5H^{\mathrm{SER}}_{\mathrm{Al}} - 0.5H^{\mathrm{SER}}_{\mathrm{Fe}}$	-12072.2	this work
	${}^{1}L^{\mathrm{B2_BCC}}_{\mathrm{Al,Fe:*:Va}} = {}^{1}L^{\mathrm{B2_BCC}}_{\mathrm{*:Al,Fe:Va}}$	-733.7	this work
	${}^{2}L^{\mathrm{B2_BCC}}_{\mathrm{Al,Fe:*:Va}} = {}^{2}L^{\mathrm{B2_BCC}}_{\mathrm{*:Al,Fe:Va}}$	-1221.9	this work
	$T^{\mathrm{B2_BCC}}_{\mathrm{CAl:Fe:Va}} = T^{\mathrm{B2_BCC}}_{\mathrm{CFe:Al:Va}}$	-360	this work
	${}^{0}L^{\mathrm{B2_BCC}}_{\mathrm{CAl,Fe:*:Va}} = {}^{0}T^{\mathrm{B2_BCC}}_{\mathrm{C*:Al,Fe:Va}}$	360	this work
	$\beta^{\mathrm{B2_BCC}}_{\mathrm{Al:Fe:Va}} = \beta^{\mathrm{B2_BCC}}_{\mathrm{Fe:Al:Va}}$	$2 \cdot -1.36$	[14]
	${}^{0}\beta^{\mathrm{B2_BCC}}_{\mathrm{Al,Fe:*:Va}} = {}^{0}\beta^{\mathrm{B2_BCC}}_{\mathrm{*:Al,Fe:Va}}$	$2 \cdot -0.3$	[14]
	${}^{1}\beta^{\mathrm{B2_BCC}}_{\mathrm{Al,Fe:*:Va}} = {}^{1}\beta^{\mathrm{B2_BCC}}_{\mathrm{*:Al,Fe:Va}}$	$2 \cdot -0.8$	[14]
	${}^{2}\beta^{\mathrm{B2_BCC}}_{\mathrm{Al,Fe:*:Va}} = {}^{2}\beta^{\mathrm{B2_BCC}}_{\mathrm{*:Al,Fe:Va}}$	$2 \cdot 0.2$	[14]

extending down to room temperature.

4. Conclusions

The heat capacity of the high-temperature ε-phase was determined by DSC measurements. The measured heat capacity is steep increasing with temperature and its values are higher than the ones reported for the other intermetallics of the system (ζ, η and θ). This steep increase may be related to the creation of thermal vacancies in the structure. However, without a proper heat capacity estimation for the metastable range, it is not possible to prove this suggestion solely on the narrow temperature range of the experiment.

Based on the latest heat capacity measurements of ζ, η and θ, a thermodynamic assessment of the Al-Fe system was done including the modelling of the homogeneity ranges of the intermetallics and the correct liquidus temperatures for the Al-rich compositions as determined in this work and reported by Refs. [6,7]. In addition, the A2/B2 ordering in the Fe-rich bcc-based solid solution was modelled reproducing the formation of the A2 + B2 two-phase field below the tricritical point.

The order/disorder transition in η [8–11], was not included in the presented modelling as more informations on the structures and phase relations in that region of the phase diagram is needed.

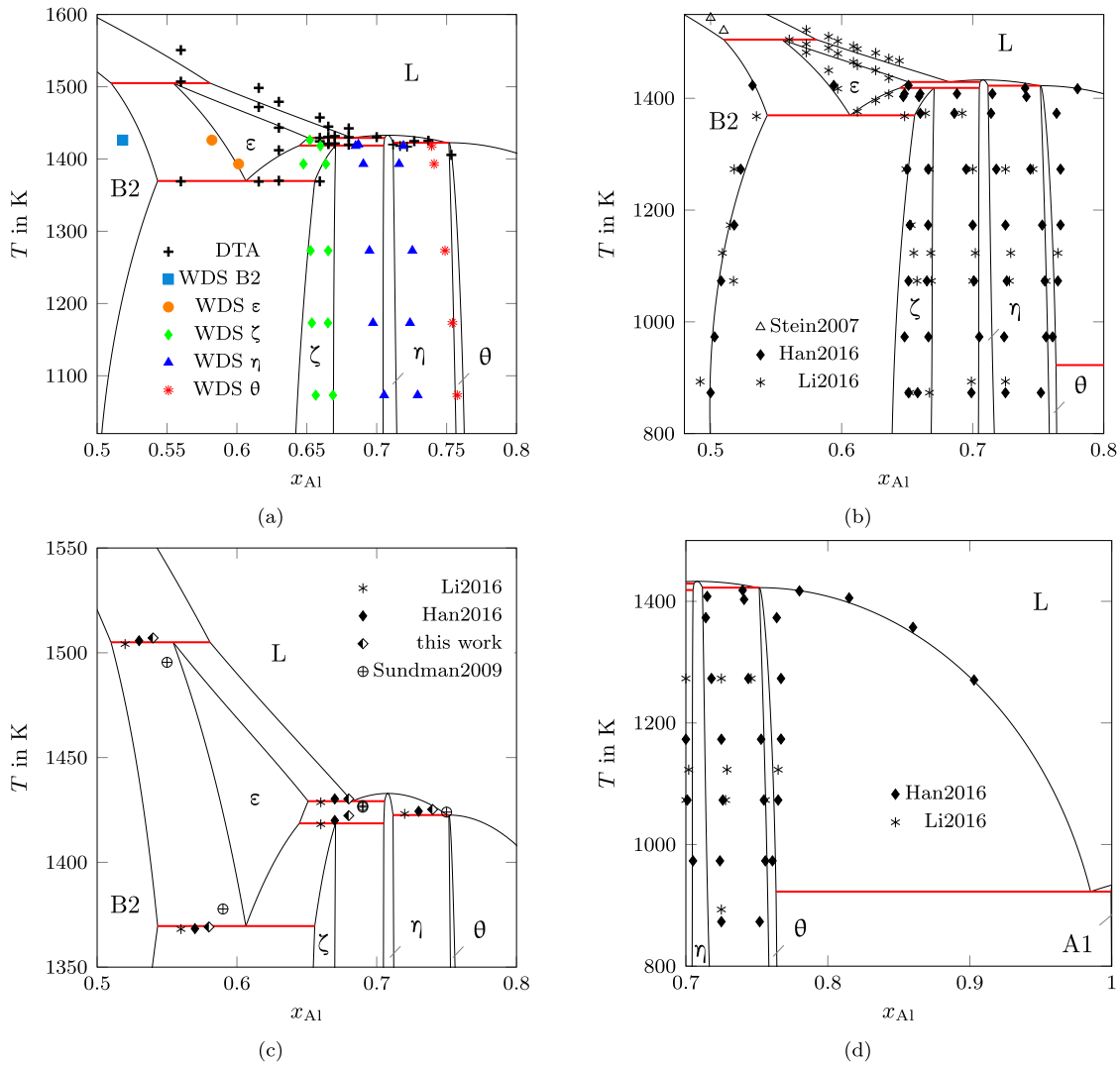

Fig. 14. Calculated phase diagrams of the Al-rich side of the Fe-Al system in comparison to experimental data of a) this work and b)+d) in comparison to reported experimental results of [6,7,27]. c) Comparison of the calculated temperatures of invariant reactions obtained in this work with reported experimental and calculated values from literature [6,7,14].

Table 5
Measured and calculated temperatures of invariant reactions of the Al-Fe system derived in this work in comparison to experimental and calculated values reported by Lee [48], Stein and Palm [27], Jacobs and Schmid-Fetzer [15], Sundman et al. [14], Han et al. [7], Li et al. [6], Phan et al. [17] and Zheng et al. [18].

Ref.	Temperature in K						
	L + B2 → ε	L → ε + η	ε + η → ζ	ε→ B2 + ζ	L → η	L + η → θ	L → θ + fcc
measured							
[48]	1501.9	1426.6	1416.7	1365.5			925.2
[27]	1504.1	1428.1	1419.1	1368.1	> 1430.1	1422.1	
[7]	1505.75	1430.25	1419.95	1368.35		1424.35	
[6]	1504.15	1428.65	1418.15	1368.15		1423.15	
this work	1507.07	1430.27	1422.28	1369.17		1425.24	
calculated							
[15], Model B	1501.2	1430.1	1428.2	1357.4	1434.7	1424.6	927.2
[14]	1495.44	1427.03	1426.29	1377.73		1423.96	
[17]	1501	1426	1417	1366		1420	926
[18]	1506	1429	1417	1370		1423	926
this work	1505.03	1429.18	1418.59	1369.58	1432.88	1422.56	922.59

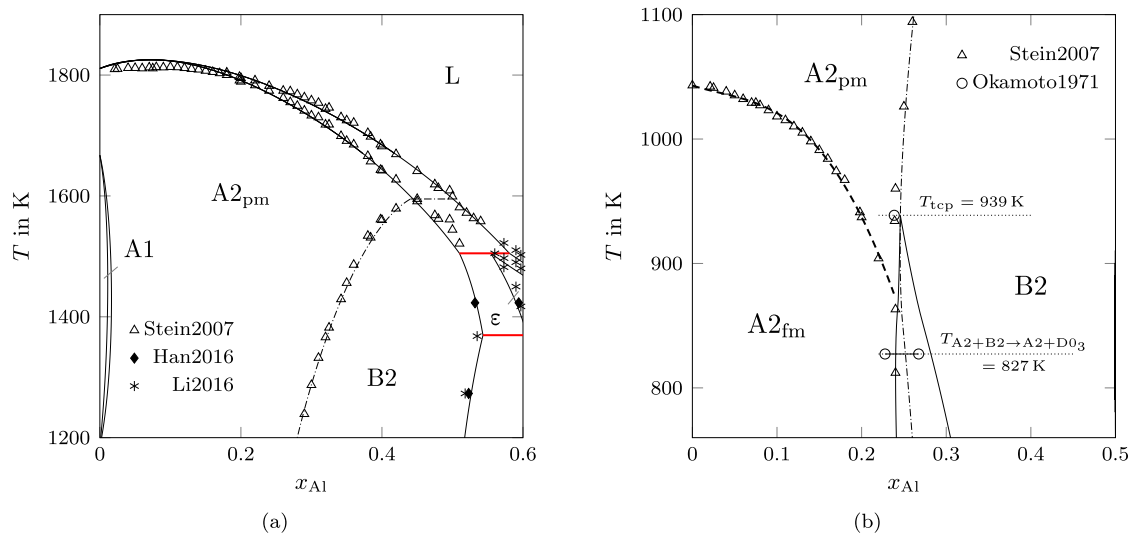

Fig. 15. a)+b) Calculated phase diagrams of the Fe-rich side of the Al-Fe system in comparison to experimental data reported by Refs. [6,7,27,46].

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Appendix A. Partial derivatives

For convenience, the magnetic thermodynamic parameters of the A2 and B2 description are abbreviated as follows

$$\begin{aligned}
&a = T^{A2}_{CFe} \quad a_0 = {}^0T^{A2}_{CAl,Fe} \quad a_1 = {}^1T^{A2}_{CAl,Fe} \quad a_2 = {}^2T^{A2}_{CAl,Fe}\\
&\beta_a = \beta^{A2}_{Fe}\\
&b = T^{B2}_{CAl:Fe} \quad b_0 = {}^0T^{B2}_{CAl,Fe:*}\\
&\beta_b = \beta^{B2}_{Al:Fe} \quad \beta_{b_0} = {}^0\beta^{B2}_{Al,Fe:*} \quad \beta_{b_1} = {}^1G^{B2}_{Al,Fe:*} \quad \beta_{b_2} = {}^2\beta^{B2}_{Al,Fe:*}
\end{aligned} \tag{A.1}$$

and T_C is given by

$$\begin{aligned}
T_C = {} & a\cdot(1-x_{Al}) + a_0\cdot\left(x_{Al} - x_{Al}^2\right) + a_1\cdot\left(-2x_{Al}^3 + 3x_{Al}^2 - x_{Al}\right)\\
& + a_2\cdot\left(-4x_{Al}^4 + 8x_{Al}^3 - 5x_{Al}^2 + x_{Al}\right) + b\cdot\left(-4x_{Al}y'_{Al} + 2x_{Al}\right.\\
& \left. + 2y'^2_{Al}\right) + b_0\cdot\left(-4x_{Al}^2 + 4x_{Al}y'_{Al} + 2x_{Al} - 2y'^2_{Al}\right)\\
& - (b+b_0)\cdot\left(-2x_{Al}^2 + 2x_{Al}\right).
\end{aligned} \tag{A.2}$$

The first and second derivative of $f(\tau)$ with respect to y'_{Al} is

$$\frac{\partial f(\tau)}{\partial y'_{Al}} = \begin{cases} -474\cdot\left(\frac{1}{p}-1\right)\cdot\frac{T^{15}\cdot K_1 + T^9\cdot K_2 + T^3\cdot K_3}{497\cdot A} - \frac{79\cdot(b_0-b)\cdot K_0}{140\cdot A\cdot T\cdot p}, & \text{for } \tau < 1\\ -\frac{\frac{(5\cdot(b_0-b)\cdot K_0)\cdot T_C^4}{10\cdot T^5} + \frac{(15\cdot(b_0-b)\cdot K_0)\cdot T_C^{14}}{315\cdot T^{15}} + \frac{(25\cdot(b_0-b)\cdot K_0)\cdot T_C^{24}}{1500\cdot T^{25}}}{A}, & \text{for } \tau \geq 1 \end{cases} \tag{A.3}$$

and

$$\frac{\partial^2 f(\tau)}{\partial y'^2_{\mathrm{Al}}} = \begin{cases} -474\cdot\left(\frac{1}{p}-1\right)\cdot\dfrac{T^{15}\cdot\frac{60\cdot(b_0-b)}{600\cdot T_C^{16}}+\frac{T^{15}\cdot 16\cdot 15\cdot K_4}{600\cdot T_C^{17}}+T^9\cdot\frac{36\cdot(b_0-b)}{135\cdot T_C^{10}}+\frac{T^9\cdot 10\cdot 9\cdot K_4}{135\cdot T_C^{11}}+T^3\cdot\frac{12\cdot(b_0-b)}{6\cdot T_C^4}+\frac{T^3\cdot 4\cdot 3\cdot K_4}{6\cdot T_C^5}}{497\cdot A} \\ -\dfrac{316\cdot(b-b_0)}{140\cdot A\cdot T\cdot p}, \quad \text{for } \tau<1 \\ -\dfrac{\frac{20\cdot(b-b_0)\cdot T_C^4}{10\cdot T^5}+\frac{4\cdot 5\cdot K_5\cdot T_C^3}{10\cdot T^5}+\frac{60\cdot(b-b_0)\cdot T_C^{14}}{315\cdot T^{15}}+\frac{14\cdot 15\cdot K_5\cdot T_C^{13}}{315\cdot T^{15}}+\frac{100\cdot(b-b_0)\cdot T_C^{24}}{1500\cdot T^{25}}+\frac{24\cdot 25\cdot K_5\cdot T_C^{23}}{1500\cdot T^{25}}}{A}, \quad \text{for } \tau\geq 1 \end{cases} \tag{A.4}$$

, respectively, with

$$\begin{aligned} &K_0 = 4x_{\mathrm{Al}} - 4y'_{\mathrm{Al}} \\ &K_1 = \frac{15\cdot(b-b_0)\cdot K_0}{600\cdot T_C^{16}} \quad K_2 = \frac{9\cdot(b-b_0)\cdot K_0}{135\cdot T_C^{10}} \quad K_3 = \frac{3\cdot(b-b_0)\cdot K_0}{6\cdot T_C^4} \\ &K_4 = (b-b_0)^2\cdot K_0^2 \quad K_5 = (b_0-b)^2\cdot K_0^2. \end{aligned} \tag{A.5}$$

The first and second derivative of $\ln(\beta+1)$ with respect to y'_{Al} is

$$\frac{\partial \ln(\beta+1)}{\partial y'_{\mathrm{Al}}} = \frac{(\beta_{b_0}-\beta_b)\cdot K_0 + K_6}{K_7} \tag{A.6}$$

and

$$\frac{\partial^2 \ln(\beta+1)}{\partial y'^2_{\mathrm{Al}}} = \frac{4\beta_b - 4\beta_{b_0} + \beta_{b_1}\cdot(-24x_{\mathrm{Al}}+12) + \beta_{b_2}\cdot\left(-192x_{\mathrm{Al}}^2 + 192x_{\mathrm{Al}}y'_{\mathrm{Al}} + 96x_{\mathrm{Al}} - 96y'^2_{\mathrm{Al}} - 20\right)}{K_7} - \frac{K_8^2}{K_7^2} \tag{A.7}$$

, respectively, with

$$\begin{aligned} &K_6 = \beta_{b_1}\cdot\left(24x_{\mathrm{Al}}^2 - 24x_{\mathrm{Al}}y'_{\mathrm{Al}} - 12x_{\mathrm{Al}} + 12y'_{\mathrm{Al}}\right) + \beta_{b_2}\cdot\left(128x_{\mathrm{Al}}^3 - 192x_{\mathrm{Al}}^2y'_{\mathrm{Al}} - 96x_{\mathrm{Al}}^2 + 96x_{\mathrm{Al}}y'^2_{\mathrm{Al}} + 96x_{\mathrm{Al}}y'_{\mathrm{Al}} + 20x_{\mathrm{Al}} - 32y'^3_{\mathrm{Al}} - 20y'_{\mathrm{Al}}\right) \\ &K_7 = \beta_a\cdot(1-x_{\mathrm{Al}}) + \beta_b\cdot\left(-4x_{\mathrm{Al}}y'_{\mathrm{Al}} + 2x_{\mathrm{Al}} + 2y'^2_{\mathrm{Al}}\right) + \beta_{b_0}\cdot\left(-4x_{\mathrm{Al}}^2 + 4x_{\mathrm{Al}}y'_{\mathrm{Al}} + 2x_{\mathrm{Al}} - 2y'^2_{\mathrm{Al}}\right) + \beta_{b_1}\cdot\left(-12x_{\mathrm{Al}}^3 + 24x_{\mathrm{Al}}^2y'_{\mathrm{Al}} + 6x_{\mathrm{Al}}^2\right. \\ &\left.-12x_{\mathrm{Al}}y'^2_{\mathrm{Al}} - 12x_{\mathrm{Al}}y'_{\mathrm{Al}} + 6y'^2_{\mathrm{Al}}\right) + \beta_{b_2}\cdot\left(-56x_{\mathrm{Al}}^4 + 128x_{\mathrm{Al}}^3y'_{\mathrm{Al}} + 48x_{\mathrm{Al}}^3 - 96x_{\mathrm{Al}}^2y'^2_{\mathrm{Al}} - 96x_{\mathrm{Al}}^2y'_{\mathrm{Al}} - 10x_{\mathrm{Al}}^2 + 32x_{\mathrm{Al}}y'^3_{\mathrm{Al}} + 48x_{\mathrm{Al}}y'^2_{\mathrm{Al}} + 20x_{\mathrm{Al}}y'_{\mathrm{Al}}\right. \\ &\left.-8y'^4_{\mathrm{Al}} - 10y'^2_{\mathrm{Al}}\right) - (\beta_b + \beta_{b_0})\cdot\left(-2x_{\mathrm{Al}}^2 + 2x_{\mathrm{Al}}\right) + 1 \\ &K_8 = (\beta_b - \beta_{b_0})\cdot K_0 - \beta_{b_1}\cdot\left(24x_{\mathrm{Al}}^2 - 24x_{\mathrm{Al}}y'_{\mathrm{Al}} - 12x_{\mathrm{Al}} + 12y'_{\mathrm{Al}}\right) - \beta_{b_2}\cdot\left(128x_{\mathrm{Al}}^3 - 192x_{\mathrm{Al}}^2y'_{\mathrm{Al}} - 96x_{\mathrm{Al}}^2 + 96x_{\mathrm{Al}}y'^2_{\mathrm{Al}} + 96x_{\mathrm{Al}}y'_{\mathrm{Al}} + 20x_{\mathrm{Al}}\right. \\ &\left.-32y'^3_{\mathrm{Al}} - 20y'_{\mathrm{Al}}\right). \end{aligned} \tag{A.8}$$

References

[1] J. Taylor, The effect of iron in Al-Si casting alloys, in: 35th Australian Foundry Institute National Conference, 2004, pp. 148–157. http://espace.library.uq.edu.au/view/UQ:100729.

[2] T. Zienert, O. Fabrichnaya, Phase relations in the A365 alloy: experimental study and thermodynamic calculations, Adv. Eng. Mater. 15 (12) (2013) 1244–1250.

[3] U. Kattner, B. Burton, Phase Diagrams of Binary Iron Alloys, ASM International, 1993, pp. 12–28. Ch. Al-Fe (Aluminium-Iron).

[4] S. Balanetskyy, B. Grushko, T. Velikanova, Monoclinic Al_2Fe phase, its equilibrium and nonequilibrium formation, Metallofiz. Noveishie Tekhnol. 26 (3) (2004) 407–417.

[5] J. Chi, X. Zheng, S.Y. Rodriguez, Y. Li, W. Gou, V. Goruganti, K. Rathnayaka, J.H. Ross, Dilute magnetism and vibrational entropy in Fe_2A_5, Phys. Rev. B 82 (2010), 174419, https://doi.org/10.1103/PhysRevB.82.174419.

[6] X. Li, A. Scherf, M. Heilmaier, F. Stein, The Al-Rich part of the Fe-Al phase diagram, J. Phase Equilibria Diffusion 37 (2) (2016) 162–173, https://doi.org/10.1007/s11669-015-0446-7.

[7] K. Han, I. Ohnuma, R. Kainuma, Experimental determination of phase equilibria of Al-rich portion in the Al-Fe binary system, J. Alloy. Comp. 668 (2016) 97–106, https://doi.org/10.1016/j.jallcom.2016.01.215. http://www.sciencedirect.com/science/article/pii/S092583881630216X.

[8] T. Zienert, L. Amirkhanyan, J. Seidel, R. Wirnata, T. Weissbach, T. Gruber, O. Fabrichnaya, J. Kortus, Heat capacity of η-AlFe (Fe_2Al_5), Intermetallics 77 (2016) 14–22.

[9] N.L. Okamoto, J. Okumura, M. Higashi, H. Inui, Crystal structure of η′-Fe_3Al_8; low-temperature phase of η-Fe^2Al_5 accompanied by an ordered arrangement of Al atoms of full occupancy in the c-axis chain sites, Acta Mater. 129 (2017) 290–299, https://doi.org/10.1016/j.actamat.2017.02.060. http://www.sciencedirect.com/science/article/pii/S1359645417301611.

[10] H. Becker, L. Amirkhanyan, J. Kortus, A. Leineweber, Powder-X-ray diffraction analysis of the crystal structure of the η′-Al_8Fe_3 (η′-$Al_{2.67}Fe$) phase, J. Alloy. Comp. 721 (Suppl C) (2017) 691–696, https://doi.org/10.1016/j.jallcom.2017.05.336. http://www.sciencedirect.com/science/article/pii/S0925838817319734.

[11] H. Becker, A. Leineweber, Atomic channel occupation in disordered η-Al_5Fe_2 and in two of its low-temperatures phases, η″ and η‴, Intermetallics (2017) https://doi.org/10.1016/j.intermet.2017.09.021 (in press), http://www.sciencedirect.com/science/article/pii/S0966979517307975.

[12] T. Zienert, A. Leineweber, O. Fabrichnaya, Heat capacity of Fe-Al intermetallics: B2-FeAl, $FeAl_2$, Fe_2Al_5 and Fe_4Al_{13}, J. Alloy. Comp. 725 (2017) 848–859, https://doi.org/10.1016/j.jallcom.2017.07.199. http://www.sciencedirect.com/science/article/pii/S0925838817325768.

[13] M. Rank, P. Gotcu, P. Franke, H.J. Seifert, Thermodynamic investigations in the Al-Fe system: heat capacity measurements of three intermetallic phases, Intermetallics 94 (2018) 73–82, https://doi.org/10.1016/j.intermet.2017.12.015. https://www.sciencedirect.com/science/article/pii/S0966979517308774.

[14] B. Sundman, I. Ohnuma, N. Dupin, U.R. Kattner, S.G. Fries, An assessment of the entire Al–Fe system including $D0_3$ ordering, Acta Mater. 57 (10) (2009) 2896–2908, https://doi.org/10.1016/j.actamat.2009.02.046. http://www.sciencedirect.com/science/article/pii/S1359645409001372.

[15] M.H. Jacobs, R. Schmid-Fetzer, Phase behavior and thermodynamic properties in the system Fe–Al, Calphad 33 (1) (2009) 170–178, https://doi.org/10.1016/j.calphad.2008.07.010. http://www.sciencedirect.com/science/article/pii/S0364591608000618.

[16] W. Gąsior, A. Dębski, Z. Moser, Formation enthalpy of intermetallic phases from Al–Fe system measured with solution calorimetric method, Intermetallics 24 (0) (2012) 99–105, https://doi.org/10.1016/j.intermet.2012.02.001. http://www.sciencedirect.com/science/article/pii/S0966979512000532.

[17] A.T. Phan, M.-K. Paek, Y.-B. Kang, Phase equilibria and thermodynamics of the Fe–Al–C system: critical evaluation, experiment and thermodynamic optimization, Acta Mater. 79 (2014) 1–15, https://doi.org/10.1016/j.actamat.2014.07.006. http://www.sciencedirect.com/science/article/pii/S1359645414005102.

[18] W. Zheng, S. He, M. Selleby, Y. He, L. Li, X.-G. Lu, J. Ågren, Thermodynamic assessment of the Al-C-Fe system, Calphad 58 (2017) 34–49, https://doi.org/10.1016/j.calphad.2017.05.003. http://www.sciencedirect.com/science/article/pii/S0364591617300482.

[19] A. Dinsdale, SGTE data for pure elements, Calphad 15 (4) (1991) 317–425, https://doi.org/10.1016/0364-5916(91)90030-N. http://www.sciencedirect.com/science/article/pii/036459169190030N.

[20] J.-O. Andersson, T. Helander, L. Höglund, P. Shi, B. Sundman, Thermo-Calc & DICTRA, computational tools for materials science, Calphad 26 (2) (2002) 273–312, https://doi.org/10.1016/S0364-5916(02)00037-8. http://www.sciencedirect.com/science/article/pii/S0364591602000378.

[21] H. Lukas, S. Fries, B. Sundman, Computational Thermodynamics - the Calphad Method, Cambridge University Press, 2007.

[22] M. Hillert, M. Jarl, A model for alloying in ferromagnetic metals, Calphad 2 (3) (1978) 227–238, https://doi.org/10.1016/0364-5916(78)90011-1. http://www.sciencedirect.com/science/article/pii/0364591678900111.

[23] G. Inden, The role of magnetism in the calculation of phase diagrams, Physica B 103 (1) (1981) 82–100, https://doi.org/10.1016/0378-4363(81)91004-4. http://www.sciencedirect.com/science/article/pii/0378436381910044.

[24] B. Sundman, J. Agren, A regular solution model for phases with several components and sublattices, suitable for computer applications, J. Phys. Chem. Solid. 42 (4) (1981) 297–301, https://doi.org/10.1016/0022-3697(81)90144-X. http://www.sciencedirect.com/science/article/pii/002236978190144X.

[25] I. Ansara, N. Dupin, H.L. Lukas, B. Sundman, Thermodynamic assessment of the Al-Ni system, J. Alloy. Comp. 247 (1) (1997) 20–30, https://doi.org/10.1016/S0925-8388(96)02652-7. http://www.sciencedirect.com/science/article/pii/S0925838896026527.

[26] N. Dupin, I. Ansara, On the sublattice formalism applied to the B2 phase, Z. Metallkd. 90 (1) (1999) 76–85.

[27] F. Stein, M. Palm, Re-determination of transition temperatures in the Fe-Al system by differential thermal analysis, Int. J. Mater. Res. 98 (2007) 580–588.

[28] M. Seierstein, COST 507-System Al-Fe, vol. 2, European Communities, 1998, pp. 34–39.

[29] R. Corby, P. Black, The structure of $FeAl_2$ by anomalous dispersion methods, Acta Crystallogr. B 29 (12) (1973) 2669–2677, https://doi.org/10.1107/S056774087300734X.

[30] I. Chumak, K.W. Richter, H. Ehrenberg, Redetermination of iron dialuminide, $FeAl_2$, Acta Crystallogr. C 66 (9) (2010) i87–i88, https://doi.org/10.1107/S0108270110033202.

[31] P. Franke, Modeling of thermal vacancies in metals within the framework of the compound energy formalism, J. Phase Equilibria Diffusion 35 (6) (2014) 780–787, https://doi.org/10.1007/s11669-014-0348-0.

[32] P.-W. Guan, Z.-K. Liu, A physical model of thermal vacancies within the CALPHAD approach, Scripta Mater. 133 (2017) 5–8, https://doi.org/10.1016/j.scriptamat.2017.02.002. http://www.sciencedirect.com/science/article/pii/S135964621730057X.

[33] U. Burkhardt, Y. Grin, M. Ellner, K. Peters, Structure refinement of the iron-aluminium phase with the approximate composition Fe_2Al_5, Acta Crystallogr. B 50 (3) (1994) 313–316, https://doi.org/10.1107/S0108768193013989.

[34] K. Schubert, U. Rösler, M. Kluge, K. Anderko, L. Härle, Kristallographische Ergebnisse an Phasen mit Durchdringungsbildung, Naturwissenschaften 40 (16) (1953) 437.

[35] K. Schubert, M. Kluge, Über den Bindungszustand bei Fe_2Al_5 und $MnAl_6$, Z. Naturforsch. 8 (1953) 755–756.

[36] M. Ellner, J. Mayer, X-ray and electron diffraction investigations on the liquid-quenched Fe_2Al_5, Scripta Metall. Mater. 26 (3) (1992) 501–504, https://doi.org/10.1016/0956-716X(92)90637-T. http://www.sciencedirect.com/science/article/pii/0956716X9290637T.

[37] P. Black, The structure of $FeAl_3$, Acta Crystallogr. 8 (1) (1955) 43–48, https://doi.org/10.1107/S0365110X5500011X.

[38] J. Grin, U. Burkhardt, M. Ellner, Refinement of the Fe_4Al_{13} structure and its relationship to the quasihomological homeotypical structures, Z. Kristallogr. 209 (1994) 479–487.

[39] F. Stein, S. Vogel, M. Eumann, M. Palm, Determination of the crystal structure of the ε phase in the Fe–Al system by high-temperature neutron diffraction, Intermetallics 18 (1) (2010) 150–156, https://doi.org/10.1016/j.intermet.2009.07.006. http://www.sciencedirect.com/science/article/pii/S0966979509001927.

[40] S.C. Vogel, F. Stein, M. Palm, Investigation of the ε phase in the Fe–Al system by high-temperature neutron diffraction, Appl. Phys. A 99 (3) (2010) 607–611, https://doi.org/10.1007/s00339-010-5619-y.

[41] O. v. Heidenstam, A. Johansson, S. Westman, A redetermination of the distribution of atoms in Cu_5Zn_8, Cu_5Cd_8, and Cu_9Al_4, Acta Chem. Scand. 22 (1968) 653–661, https://doi.org/10.3891/acta.chem.scand.22-0653.

[42] O. Gourdon, D. Gout, D.J. Williams, T. Proffen, S. Hobbs, G.J. Miller, Atomic distributions in the γ-brass structure of the Cu-Zn system: a structural and theoretical study, Inorg. Chem. 46 (1) (2007) 251–260, https://doi.org/10.1021/ic0616380 pMID: 17198434. arXiv.

[43] F. Woolley, J.F. Elliott, Heats of solution of aluminium, copper, and silicon in liquid iron, Trans. Metall. Soc. AIME 239 (1967) 1872–1883.

[44] M. Petrushevskiy, Y. Esin, P. Gel'd, V. Sandakov, Concentration dependence of the enthalpy of formation of liquid iron-aluminium alloys, Russ. Metall. 6 (1972) 149–153.

[45] P. Swann, R. Fisher, Effect of magnetization on the $L2_0$ ordering reaction in iron-aluminum alloys, Appl. Phys. Lett. 9 (8) (1966) 279–281, https://doi.org/10.1063/1.1754749. http://link.aip.org/link/?APL/9/279/1.

[46] H. Okamoto, P.A. Beck, Phase relationships in the iron-rich Fe-Al alloys, Mater. Trans. 2 (2) (1971) 569–574, https://doi.org/10.1007/BF02663350.

[47] S.M. Allen, J.W. Cahn, Coherent and incoherent equilibria in iron-rich iron-aluminum alloys, Acta Metall. 23 (9) (1975) 1017–1026, https://doi.org/10.1016/0001-6160(75)90106-6. http://www.sciencedirect.com/science/article/pii/0001616075901066.

[48] J. Lee, Liquidus-solidus relations in the system iron-aluminium, J. Iron Steel Inst. 194 (1960) 222–224.